

Influence of biological sex in inflammatory bowel diseases

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Abstract

The incidence of inflammatory bowel disease (IBD) has risen in the past decades and has emerged as a global health issue. IBD is characterized by chronic inflammation of the gastrointestinal tract. There is compelling evidence for the role of biological sex in IBD epidemiology, pathophysiology, disease progression, symptoms and extra-intestinal diseases. IBD disease course, management and therapies differ between men and women, yet there is a paucity of analysis of sex as a factor. This Review discusses the known influence of sex-linked genetic factors, hormones and hormone receptors in IBD incidence, prevalence, disease burden and clinical manifestation. Furthermore, we review the mechanisms underlying these sex-dependent effects on the dysregulation of gastrointestinal mucosal immunity (immune, epithelial barrier and microbiota) in IBD. To support the progressive inclusion of sex in the study of IBD, we summarize the current standard research methodology that should be implemented to investigate sex as a biological variable in IBD studies. Enhanced comprehension of the influence of sex in IBD pathophysiology will advance the development of targeted therapies and improve patient care.

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Key points

- Biological sex differentially affects inflammatory bowel disease (IBD) subtypes, ulcerative colitis, and Crohn's disease epidemiology, disease course and symptoms across ages.
- This article summarizes current knowledge of the effect of biological sex on IBD, highlights research gaps and recommends best practices in research methodology to integrate sex in IBD research.
- To improve patient care, integration and comprehension of biological sex in IBD, personalized therapeutic strategies and the inclusion of a broader population in preclinical and clinical research are needed.

Introduction

Historically, biomedical research has lacked adequate representation of biological sex in preclinical and clinical research despite a growing body of evidence highlighting the role of sex differences in disease susceptibility, therapeutic strategies and clinical outcomes^{1–9}. In current years, awareness of this issue has risen and led funding agencies and scientific journals to promote policies that require researchers to consider sex as a biological determinant to implement in the study design, data collection and analysis, and reporting of all preclinical and clinical research^{10–12}.

Inflammatory bowel diseases (IBD) include Crohn's disease, ulcerative colitis and IBD unclassified (IBDU), which are chronic and severely debilitating gastrointestinal diseases of unknown aetiology. Clinical studies have reported modest sex associations in IBD epidemiology, disease progression, symptoms and response to therapy^{4,5,13,14}. Yet, the precise role of sex in IBD remains poorly understood^{4,5,13,14}. To better understand the effects of biological sex in IBD, research teams need to appropriately integrate sex considerations into study design, supporting the discovery of personalized therapeutic strategies^{6,10}. This Review discusses the current knowledge regarding the involvement of sex-related dysmorphism in IBD, outlining the potential underlying mechanisms and biases or differences associated with IBD pathology. Lastly, we discuss the current best practices of sex research methodology from study design, participant selection and data collection to analysis, interpretation, and data presentation.

Defining sex and gender

With emerging evidence suggesting that sex and gender contribute to differences in IBD, it becomes imperative to define the terms 'sex' and 'gender' and to recognize that these definitions continue to evolve^{4,5,13–15}. Sex describes the biological attributes that account for physical and physiological characteristics of men, women, intersex and individuals undergoing sex reassignment. Gender refers to socially constructed roles encompassing behaviours, identities and expressions^{16,17}. Thus, biological sex is defined as the assigned sex, male or female, given to an individual at birth based on their biological sex attributes, including sex hormones, chromosomes, reproductive organs and sexual anatomy. Typically, humans have 46 chromosomes, of which 2 are sex chromosomes X and Y, which encode for individual biological sex features associated with the assigned sex at birth¹⁸. In that context, the combination of two X chromosomes typically led to a female phenotype associated with the development of ovaries, which produce sex hormones that regulate secondary sex determinants such as oviducts,

uterus, cervix, clitoris, labia, vagina and mammary glands. Meanwhile, an individual identified as male typically has one X and one Y chromosome and develops testes that produce sex hormones that promote the development of seminal vesicles and prostate gland¹⁸. However, chromosomal, gonadal or anatomical atypical development, termed disorder of sex development, can result in a mismatch between an individual chromosomal pattern, sex hormone profile and reproductive system, leading to the birth of an intersex individual with sex biological traits that do not fit the dichotomic notions of female or male classification¹⁹. Individuals with intersex traits, estimated to account for 0.018–1.7% of the global population, can be identified at birth owing to atypical sexual anatomy or later affected by postnatal virilization, delay or absence of puberty, or infertility^{19–21}. Overall, these definitions emphasize the large spectrum associated with biological sex, increasing the complexity of investigating the sex-mediated difference in the biomedical field. Although complex, the attributes of biological sex are relatively well defined, therefore facilitating the investigation of their effect on IBD.

By contrast, the notion of gender is a more complex and fluctuating social construct that differs with culture, environment, age and era, preventing the development of standardized gender scoring systems and limiting their general application to broad IBD research¹⁶. As a multifaceted and fluid construct, the concept of gender is influenced by societal, cultural and environmental factors to create gender norms that, in turn, affect individual behaviour, interaction with others, perception of oneself and others, as well as gender distribution of power and resources in society²². In that context, gender is often categorized along four constructs, including gender role, gender identity, gender relation and institutionalized gender²². Biological sex can affect the disease pathophysiology, manifestations and response to treatment, whereas gender influences health care-seeking behaviours, decision-making, and therapeutic response²³. Thus, poor integration of sex and gender in biomedical research can lead to false interpretation, misdiagnosis, delay in treatment and increased risk for adverse drug reactions^{23,24}.

Sex factors on IBD clinical manifestation

Social determinants of health in IBD

Social determinants of health (SDOH) include socio-economic factors, the health-care system, personal behaviours and environmental conditions. Subdivided into intermediary SDOH, broadly categorized into food security, housing, transportation access, psychosocial and socio-economic factors, SDOH is a determining factor that causes intrinsic societal stratification, creating and sustaining inequality across populations and shaping an individual and population health status^{25,26}. According to a population-based nationwide survey conducted in the USA, 42% of patients with IBD reported being affected by food insecurity and reduced social support^{27,28}. In a population-based study from Canada using health administrative-based data with a cohort of 9,298 individuals with IBD, lower socio-economic status was linked to worsened health outcomes such as increased health care utilization and death for individuals with IBD^{27,28}. In a separate retrospective study conducted in the USA using electronic medical records from 175 patients with IBD, increased risk for IBD-related hospitalization and surgery was associated with food insecurity, financial resources and transportation, which affected more patients with Crohn's disease than with ulcerative colitis (area under the receiver operating characteristic (AUROC): 0.86 and 0.68 for Crohn's disease and ulcerative colitis, respectively), confirming the strong influence of SDOH in

IBD outcome²⁹. Biological sex and gender interplay with other SDOH, influencing the differential exposure to intermediary SDOH, which, in turn, affects the risk of exposure, use and access to health-care services and bias in health-care systems, further shaping the development and implementation of health policies^{29,30}. Thus, sex and gender, along with other SDOH, have a strong influence on an individual or a community's experience of illness and health outcomes. Yet, data on sex and gender interaction with other SDOH in IBD outcomes are limited²⁷.

Sex differences in lifestyle behaviours in IBD

Numerous environmental factors, including lifestyle, pharmacological agents and ecological determinants, are sex-specific and gender-specific and likely to contribute to IBD pathogenesis and severity. Men and women have considerable differences in substance exposure and lifestyle risk factors^{31–33}. Epidemiological and clinical studies strongly correlated poor diet, excessive alcohol consumption, smoking, and physical inactivity with a higher risk for chronic non-communicable diseases, such as cardiovascular diseases, type 2 diabetes, cancer and IBD, and are strongly influenced by SDOH³³.

Studies across several countries worldwide report that women consumed more vegetables and fruit than men^{33–36}. By contrast, a few of these studies described a higher fat and meat consumption by men than by women^{35,36}. Evidence associated dietary behaviours with IBD onset and disease severity: high consumption of dietary fibres from fruits and vegetables has been associated with a lower risk of developing Crohn's disease with no substantial changes in the risk of ulcerative colitis³⁷. Conversely, high levels of total fat, saturated fat, polyunsaturated fatty acid and omega-6 fatty acid intake increased the risk of developing ulcerative colitis and Crohn's disease. Some studies similarly reported an association between a high intake of meat and an increased risk for IBD³⁷. Despite the contradiction, several studies reported differences in macronutrient intake between patients with different IBD subtypes compared to the general population³⁸. Thus, patients with IBD had lower fibre consumption compared to healthy controls. When stratified by subtypes, a Dutch case-control study analysing the dietary behaviours of 207 patients with ulcerative colitis stratified by sex reported a lower intake of total proteins, animal proteins, carbohydrates and alcohol but higher fat consumption compared to control groups³⁹. These results were partially in line with several other studies^{40,41}. Similarly, patients with Crohn's disease had a lower intake of total protein, animal-based proteins and alcohol³⁹. However, unlike in ulcerative colitis, the cohort of 286 patients with Crohn's disease consumed less plant-based proteins and reported a comparable fat intake to the control groups³⁹. The previously described Dutch case-control study of 493 patients with IBD reported negligible differences in dietary intake between men and women, in concordance with a German monocentric cross-sectional study with a cohort of 82 patients with IBD and an age-matched cohort of 328 individuals representing the general German population, which described minor sex-related differences in the dietary behaviours of patients with IBD^{39,42}. By contrast, a systematic review with a meta-analysis found a correlation between dietary protein intake and IBD risk in women but not in men in the Asian population from case-control studies performed in Iran, Israel and Japan⁴³. However, the study reported limited differences in IBD risk associated with elevated protein consumption in the European and North American populations⁴³. In a cross-sectional population-based study conducted in the USA with a cohort of 3 million individuals with IBD, men with IBD were more likely to consume vegetables once or more per day compared to men

without IBD (84.9% (95% CI 77.6–90.1) versus 76.0% (95% CI 75.0–77.0); $P < 0.01$) and take any supplements (59.6% (95% CI 46.8–71.3) versus 46.0% (95% CI 44.8–47.2); $P < 0.05$) whereas women with IBD were more likely to consume sugar-sweetened beverages twice or more per day (26.8% (95% CI 19.7–35.3) versus 17.8% (95% CI 17.0–18.7); $P < 0.05$) and take vitamin D supplementation (37.2% (95% CI 27.9–47.5) versus 23.6% (95% CI 22.8–24.5); $P < 0.01$) when compared to women without IBD⁴⁴. No significant differences in the intake of other vitamins were observed between women with and without IBD. These findings highlight the need to investigate and stratify data in macronutrient sources, dietary patterns and geographical determinants.

According to the most current WHO 2019 report, an estimated 2.5 billion individuals aged 15 years and older consume alcohol worldwide⁴⁵. Although sex differences in alcohol consumption are universal, with a global prevalence rate of 52.2% for men and 35.4% for women, these differences vary markedly across geographical regions, countries and cultures, with men-to-women ratios of current drinkers ranging, for example, from 1.29 in the European Region to 2.68 in the Eastern Mediterranean Region^{45,46}. Moreover, current studies reported a rise in alcohol consumption for women but not for men, suggesting a narrowing in alcohol use gender gaps^{47,48}. However, women are more susceptible to alcohol use disorder, alcohol-related harms (memory blackout and hangover) and alcohol-associated medical consequences (alcohol-induced liver inflammation, cardiomyopathy and breast cancer) than men^{47–49}. Although controversial due to insufficient and contradictory evidence, alcohol consumption might affect IBD pathophysiology, increase the risk of relapse, and enhance gastrointestinal symptoms potentially by altering the microbiota and promoting intestinal inflammation^{50,51}. Moreover, in vitro studies and clinical studies conducted on patients with autoimmune conditions suggested that alcohol consumption could interfere with the metabolism of IBD medications such as cyclosporin, methotrexate and 5-aminosalicylic acid by decreasing their efficacy or promoting liver disease^{50,51}. However, our understanding of the interplay between sex, alcohol consumption and IBD is lacking and will require further investigation.

The Global Burden of Disease, Injuries and Risk Factors Study reported that the proportion of adult smoking differs markedly between sexes, with a higher prevalence of smokers amongst men than amongst women (32.6% (95% CI 32.2–33.1) and 6.5% (95% CI 6.3–6.7), respectively)⁵². Tobacco smoking is the earliest and most extensively studied environmental risk factor that correlates with IBD natural history. In the past 40 years, several clinical studies have consistently reported a dichotomous relationship between smoking and the onset of IBD subtypes, as reports showed that smoking increased the risk of developing Crohn's disease (relative risk estimate (RR) 1.58, 95% CI 1.37–1.82; $P < 0.001$) but has a protective effect in ulcerative colitis (RR 0.55, 95% CI 0.48–0.64; $P < 0.001$)⁵³. Yet, when stratified by biological sex, all data suggest that women with IBD are more susceptible than men to the harmful effects of smoking. A case-control study conducted in Sweden between 1984 and 1987 with 552 patients with IBD reported a fivefold increased risk for women of developing Crohn's disease in people who smoked compared to their non-smoker counterparts (RR 4.99, 95% CI 2.7–9.2; $P < 0.001$) and unlike the 30% increased risk for Crohn's disease observed in men currently smoking (RR 1.33, 95% CI 0.7–2.6; $P < 0.001$)⁵⁴. In a population study of 1,784 patients with IBD, smoking was also associated with an increased number of women with Crohn's disease diagnosed with more severe disease presentation (pancolitis: 46% compared to 32%; $P < 0.01$) and an increased need for immunosuppressive therapy in people who smoked compared to

non-smokers ($58 \pm 4\%$ compared to $48 \pm 5\%$; $P < 0.01$), whereas smoking did not significantly affect the disease course and severity in men with Crohn's disease⁵⁵. Moreover, in a small case-control study with 55 adults with Crohn's disease, the increased susceptibility to smoking was associated with higher levels of pro-inflammatory cytokines and decreased levels of regulatory T (T_{reg}) cells in women compared to men who smoked ($P = 0.05$)⁵⁶. These findings corroborate the susceptibility of women with Crohn's disease to smoking and are consistent with observations from other studies^{57,58}. However, the effect of smoking on ulcerative colitis is unclear, as studies reported that women who currently smoked were less likely to develop ulcerative colitis compared to people who did not smoke^{54,59,60}. Other studies found no significant differences in the development of ulcerative colitis between people who currently smoked and those who did not^{57,61}. The current findings result from studies conducted, for the most part, in a non-Jewish white population⁶². Given the differential association between smoking in IBD and ethnicity, more studies will help to characterize the interaction between tobacco smoking in IBD and biological sex in other ethnic groups⁶². Moreover, further studies will be required to clarify the underlying mechanisms that regulate the susceptibility of women with Crohn's disease and ulcerative colitis to smoking.

The WHO considers insufficient physical activity as an emerging public health issue associated with increased risk factors for obesity, non-communicable diseases, mental health issues and overall poor quality of life⁶³. In a population-based global survey conducted in 2016, 27.5% (95% uncertainty interval (UI) 25.0–32.2) of adults and 19% (95% UI 12.3–22.2) of adolescents had insufficient physical activity^{64,65}. Moreover, across nearly all countries, girls and women (girls: 84.7%, 95% UI 83.0–88.2; women: 31.7%, 95% UI 28.6–39.0) were less active than boys and men (boys: 77.6%, 95% UI 76.1–80.4; men: 23.4%, 95% UI 21.1–30.7) with differences of prevalence of 7.1% for adolescents and 8.3% for adults^{64,65}. Despite the scarcity of studies investigating the effect of physical activity on IBD, most studies correlate physical activity positively with IBD onset and disease course. Several case-control and retrospective studies found that sedentary occupations increased the risk of developing IBD in men and women while increasing hospitalization and mortality rates in patients with IBD^{66–68}. Additional case-control and prospective cohort studies associated infrequent physical activity in childhood with an elevated risk of subsequently developing Crohn's disease and ulcerative colitis in adulthood⁶⁹. By contrast, increased physical activity in adulthood correlated with a reduced likelihood of developing Crohn's disease but not ulcerative colitis, regardless of ethnicity⁷⁰. Moreover, amongst patients with IBD, a higher level of physical activity correlated with a lower risk of active Crohn's disease, less extended ulcerative colitis, and reduced risk of relapse at 6 months in Crohn's disease and ulcerative colitis^{71,72}. These findings were consistent with two population-based prospective studies conducted with a cohort of 121,700 women registered nurses in the USA and 240,984 Swedish men who underwent compulsory military conscription⁷³. Women who partake in vigorous physical activity had a lower risk of developing Crohn's disease than the less active women (HR 0.63, 95% CI 0.42–0.95; $P = 0.02$)⁷³. However, physical activity did not correlate with the likelihood of developing ulcerative colitis in women⁷³. Interestingly, in men, low fitness correlated with increased risk for Crohn's disease (HR 1.62, 95% CI 1.31–2.00) and ulcerative colitis (HR 1.36, 95% CI 1.17–1.59). Overall, these findings suggest a protective role for physical activity in IBD regardless of biological sex⁷⁴. Further investigations are needed to characterize the mechanisms associated with the protective role of physical activity.

Sex differences in IBD epidemiology

In the past decade, the incidence and prevalence of IBD have steadily risen, becoming an emerging global health issue that affects 4.90 million individuals worldwide⁷⁵. Even though there are conflicting results, mounting evidence points to sex biases in the epidemiology, pathogenesis, pathophysiology, disease course and clinical outcomes of IBD^{4,5,13–15}. Sex bias refers to the underreporting or misreporting of demographic, social and economic characteristics associated with one of the sexes. It is worth noting that sex differences in IBD seem to be an age-dependent relationship, further affected by factors such as geographical location^{4,13}. Moreover, the effect of sex differs between Crohn's disease and ulcerative colitis^{4,13}. Although IBD can develop at any age, the most common age for IBD diagnosis ranges from 10 to 30 years, accounting for >25% of all reported IBD cases⁷⁶. The known influence of sex in IBD epidemiology across age groups is visualized in Fig. 1.

Sex differences in very-early-onset IBD epidemiology

Very-early-onset (VEO) IBD (VEO-IBD) is a subset of paediatric IBD (pIBD) in which patients are diagnosed before 6 years of age^{77,78}. VEO-IBD accounts for 3–15.4% of all pIBD diagnoses and is sub-categorized by age into infantile-onset (<2 years) and neonatal-onset (by 28 days) IBD^{77,79,80}. Infantile-onset IBD is rare and equally distributed between Crohn's disease (36%), ulcerative colitis (31%) and IBDU (33%)^{79–81}. A multi-centre registry-based study with 1,370 patients reported that the proportion of diagnosed IBDU cases decreased markedly with age at diagnosis to -18% for older paediatric patients (3–17 years) and 6% for adults⁷⁹. Ulcerative colitis is the most predominant form of IBD diagnoses between 3 and 5 years (47% versus 35% for Crohn's disease and 18% for IBDU), contrasting with the higher prevalence of Crohn's disease (60–66%) in older paediatric patient age groups⁷⁹.

Although there is a higher prevalence of boys being diagnosed with infantile-onset IBD (63%) compared to girls, studies have not identified a significant difference in the distribution of sexes within each VEO-IBD age group^{79,80}. However, a Canadian population-based retrospective study with a cohort of 7,143 paediatric patients investigating the incidence and outcomes of pIBD suggested differences in clinical outcomes between boys and girls in the VEO-IBD age group⁸². The researchers found that girls in the VEO-IBD age groups had less outpatient physician visits (odds ratio (OR) 0.70, 95% CI 0.50–0.98 Crohn's disease; 0.75, 95% CI 0.57–0.97 ulcerative colitis) and lower hospitalization rates (Crohn's disease HR 0.56, 95% CI 0.37–0.85; $P < 0.006$) compared to boys (HR 1.07, 95% CI 0.81–1.41; $P < 0.65$). By contrast, boys diagnosed with ulcerative colitis in the VEO-IBD age groups had a lower risk of undergoing a colectomy (HR 0.42, 95% CI 0.21–0.85) compared with girls in the same age group (HR 0.88, 95% CI 0.47–1.63)⁸². The limited number of VEO-IBD cases might hinder the investigation of the effect of sex within that population, highlighting the importance of conducting more extensive multi-centre and international cohort studies. However, when designing these studies, it is essential that teams recognize that environmental factors and geographical location also seem to influence sex associations in IBD^{4,83,84}.

Sex differences in IBD epidemiology in children and adolescents

Roughly 10–25% of all patients with IBD are diagnosed before 20 years of age, although the global paediatric incidence of IBD varies greatly between geographical locations^{79,85–89}. Most systematic studies since 1950 have reported rapid growth in the incidence and prevalence of pIBD^{85,89}. Age-stratified studies of Western and Korean patients with

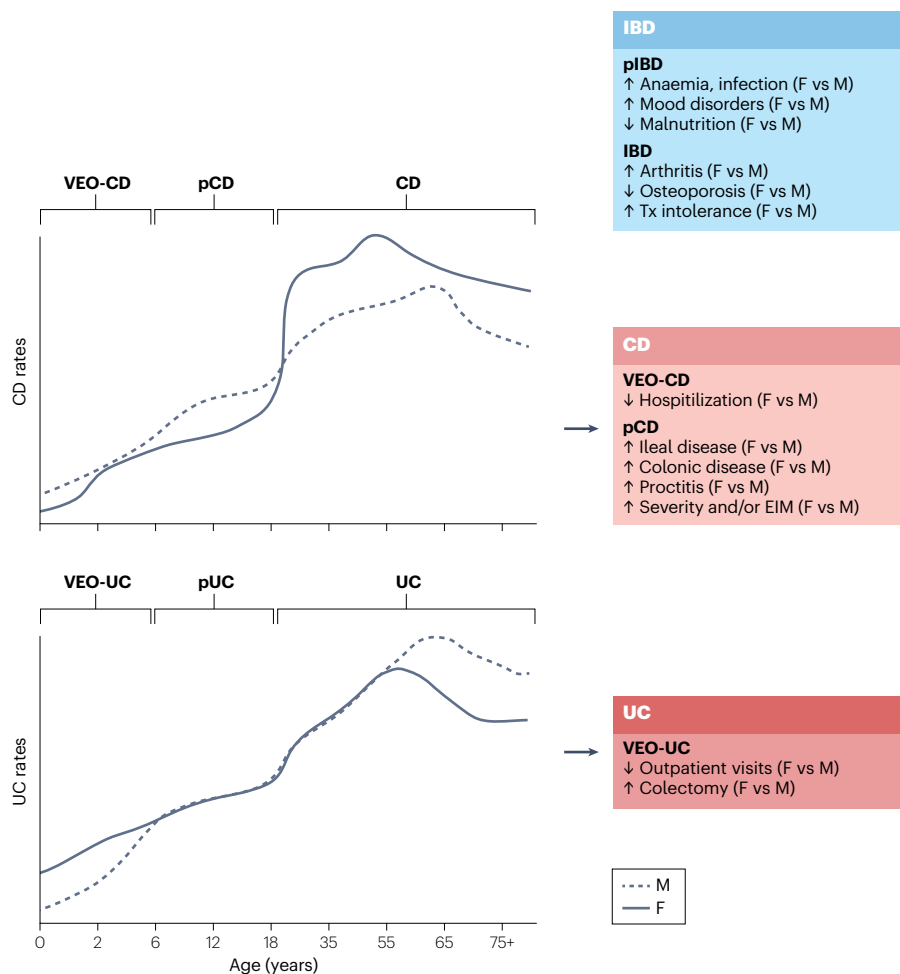


Fig. 1 | Biological sex differences in IBD epidemiology across age groups. Female (F) and male (M) patients show substantial differences in Crohn's disease (CD) and ulcerative colitis (UC) incidence and outcomes across ages from very-early-onset (VEO) and paediatric to adult CD and UC onset. Plotted values are hypothetical and for illustrative purposes. EIM, extra-intestinal manifestations; IBD, inflammatory bowel disease; pCD, paediatric CD; pIBD, paediatric IBD; pUC, paediatric UC; Tx, treatment.

IBD showed that the highest prevalence of pIBD occurred during adolescence (10–17.9 years)^{87,89–93}. However, the age of referral from paediatric to adult care for IBD management differs substantially within various literature sources, therefore making the comparison of studies of patients with pIBD across diverse populations challenging⁸⁸.

Despite regional variability, the prevalence and incidence of Crohn's disease and ulcerative colitis consistently differ markedly between paediatric boys and girls⁹⁴. Large population-based studies in European, North American and East Asian paediatric populations reported a boys predominance in Crohn's disease prevalence (male-to-female ratio 1.34–6.5:1) and incidence (male-to-female ratio 1.32–1.91:1) compared to girls^{87,89,90,93,95–97}. By contrast, the incidence and prevalence of ulcerative colitis by biological sex in the paediatric population are ambiguous. Most studies reported a similar rate of paediatric ulcerative colitis between boys and girls^{4,95,98}. A retrospective population-based German study of pIBD in a cohort of 712 patients reported a similar incidence (male-to-female ratio 0.95:1) and a lower prevalence of boys with ulcerative colitis to girls (male-to-female ratio 0.7:1)⁹⁰. Subsequently, a Western study with a cohort of 236,181,614 men and 242,373,419 women found an increased incidence of ulcerative colitis in girls aged 5–9 years with a female-to-male ratio of 1.22:1 (ref. 4), supported by a multi-centre study with a cohort of 196 patients with IBD from Switzerland, which reported increased paediatric ulcerative

colitis predominance in girls (female-to-male ratio 1:0.78)⁹⁶. By contrast, studies in Asian-Pacific paediatric populations reported a significantly ($P < 0.05$) increased incidence of ulcerative colitis in boys (male-to-female ratio 3.1:1) compared to girls, beginning in mid-late adolescence (>15 –19 years)^{83,87}. Factors including regional variability, relatively small cohort sizes and variability in the stratification of paediatric age groups might explain the discrepancy noted in the conflicting studies^{88,93}.

Sex bias in pIBD extra-intestinal clinical manifestation

Cross-sectional studies using an administrative data base from 44 affiliated children's hospitals with a cohort of 5,782 hospitalized patients with pIBD in the USA showed that adolescent girls are more likely to have anaemia (29% versus 25%; $P = 0.012$), hypoalbuminaemia (75% versus 57%; $P = 0.01$), mood disorders (9% versus 6%; $P < 0.001$) and infection (12% versus 8%; $P = 0.001$) than boys. However, adolescent boys with pIBD are more affected by delays in maturation, including growth and sexual development, than girls (3% versus 1%; $P < 0.001$)^{76,97,99}. One report with a cohort of 5,782 adolescents with Crohn's disease suggested that boys have greater rates of malnutrition than girls with pIBD (18% versus 14%; $P = 0.027$), although the investigators did not define malnutrition in that report⁹⁷. In individuals who experience IBD onset before puberty, studies with a cohort of 87 and 34 individuals

reported that the disease was commonly accompanied by delayed puberty, although this occurred more frequently in Crohn's disease than in ulcerative colitis^{100,101}. Hence, paediatric patients with Crohn's disease were more likely than patients with ulcerative colitis to present with delays in bone development, testicular enlargement, pubertal growth spurts, breast development and menarche^{100–102}.

Sex differences in IBD epidemiology in adults

Most commonly diagnosed between the ages of 15 and 30 years, the prevalence of IBD increases with age and peaks at age 50–54 years for women and 60–64 years for men⁷⁵. Comprehensive analyses of the global IBD trend revealed that women have a higher prevalence of IBD than men, accounting for 57% of all cases^{75,103}. Despite the discrepancy in the reported results, most studies described a sex bias in the epidemiology of Crohn's disease and, to a lesser extent, ulcerative colitis¹⁰⁴. Investigations in Western countries observed a predominance of adult Crohn's disease in women compared to men (female-to-male ratio 1.20:1)^{4,105,106}, with a shift in the sex ratio of Crohn's disease incidence from male predominance (incidence rate ratio (IRR) for women to men 0.83–0.70:1) from ages 5 to 14 years to a female predominance (IRR 1.27–1.38:1) from 25 to 75 years⁴. On the other hand, epidemiology surveys across Southeast Asian and Western-Pacific regions reported a higher prevalence of Crohn's disease in male than in female patients (female-to-male ratio 0.5:1) in all age ranges from 10 to 50 years^{83,98,104}.

Several European, American and Asian population studies reported a predominance of ulcerative colitis in men (women-to-men IRR 0.74–0.8:1) but data are inconsistent with other studies demonstrating no significant difference^{4,80,83,98,107–111}. By contrast, studies in Macau with a cohort of 500,000 individuals and Hungary found a higher incidence rate of ulcerative colitis in women than in men^{4,112}. Taking age into account, the incidence of ulcerative colitis was comparable between both sexes from 10 to 45 years (IRR 1.01–1.07:1) in patients with IBD from Western regions; however, the prevalence was higher in men than in women (IRR 0.83–0.92:1) after 45 years⁴. Similar results were reported in Asian-Pacific patients with ulcerative colitis, indicating a higher incidence of ulcerative colitis in male individuals between the ages of 15 and 65 years^{13,83,98}.

The shift in the biological sex ratio of Crohn's disease incidence from men to women predominance and of ulcerative colitis from women to men predominance occurs around puberty and menopause, hinting towards a role for sex hormones in the pathogenesis of both Crohn's disease and ulcerative colitis⁴. Moreover, the sex distribution of IBD epidemiology differs markedly between different geographical regions, suggesting a potential role for environmental exposure in sex-mediated IBD pathogenesis^{4,83,98}. Consequently, this discrepancy underscores the need to investigate unexplored countries and territories worldwide, focusing on well-powered cohort studies.

Sex bias in adult IBD extra-intestinal clinical manifestations

Extra-intestinal clinical manifestations and complications, a hallmark of IBD, can affect the musculoskeletal, ocular, cutaneous, hepatobiliary, renal, pulmonary, cardiovascular and haematological systems¹¹³. According to several clinical studies, the prevalence of extra-intestinal clinical manifestations in patients with IBD ranges from 6% to 47%, although these prevalences differ in a sex-dependent manner^{104,106,113–116}. Thus, a Dutch IBD biobank study with a cohort of 2,118 patients with Crohn's disease and 1,269 patients with ulcerative colitis reported that extra-intestinal clinical manifestations are more common amongst women with IBD than amongst men with IBD¹⁰⁶.

Accordingly, cutaneous and musculoskeletal manifestations (arthritis and arthropathy) affect substantially more women with Crohn's disease and ulcerative colitis than men with both IBD subtypes. Conversely, men with Crohn's disease have a higher prevalence of osteopenia than women with Crohn's disease ($P < 0.01$)¹⁰⁶. A population-based longitudinal study with a cohort of 4,454 patients with IBD for 10 years or more found an increased association between Crohn's disease and the prevalence of erythema nodosum (1.9–7.2%)¹¹⁴. The same study found that primary sclerosing cholangitis and pyoderma gangrenosum are more prevalent in ulcerative colitis than Crohn's disease. Iritis and uveitis had similar rates between Crohn's disease and ulcerative colitis¹¹⁴. When stratified by biological sex, women with ulcerative colitis were more likely to develop iritis or uveitis (3.2% in women versus 0.9% in men; $P = 0.01$) and erythema nodosum (1.9% in women versus 0.7% in men, $P = 0.02$)¹¹⁴. By contrast, men with ulcerative colitis were more affected by primary sclerosing cholangitis compared to women with ulcerative colitis and men with Crohn's disease, whereas ankylosing spondylitis is more common in men with Crohn's disease than in men with ulcerative colitis and in women with either of both IBD subtypes¹¹⁴.

Meta-analyses and population-based studies reported sex-based differences associated with IBD-mediated complications such as cancer and cardiovascular disorders¹¹³. Thus, a large population-based Swedish study with a cohort of 7,607 patients with IBD reported that both men and women with IBD had a higher risk of developing colorectal cancer compared to the general population (men: RR 2.6, 95% CI 2.2–3.1; women: RR 1.9, 95% CI 1.5–2.4)¹¹⁷. However, men with IBD had a 60% higher risk of developing colorectal cancer than women with IBD (cumulative incidence 8.3% versus 3.5%)^{113,117}. Conversely, several studies reported that patients with IBD have an increased risk for cardiovascular disease compared to the general population^{113,118,119}. Moreover, cardiovascular diseases are more common amongst women with IBD than amongst men with IBD^{113,118,119}.

Differences between sexes can also occur in response to IBD therapies such as corticosteroids, anti-inflammatory drugs, immunomodulators and biologics (TNF blockers)^{8,120}. Cross-sectional studies and retrospective cohort studies have shown that women with IBD had an increased risk of developing intolerance to azathioprine (RR 1.56; $P = 0.038$)^{76,121}. Moreover, women with IBD had increased hypersensitivity reactions to the biologic infliximab, reduced responsiveness to the biologic adalimumab and increased side effects to TNF blockers, resulting in lower drug persistence to biologic therapy for women and a greater shift to corticosteroid treatment^{76,122}. Furthermore, a meta-analysis of observational studies on vedolizumab effectiveness and safety reported an increased clinical remission rate at maintenance (OR 1.021, 95% CI 1.003–1.040; $P = 0.043$) and induction (OR 1.027, 95% CI 1.006–1.049; $P = 0.015$) and a higher clinical response rate (OR 1.025, 95% CI 1.006–1.046; $P = 0.0022$) for men with Crohn's disease but no significant difference between sexes in ulcerative colitis¹²⁰.

Sex differences in IBD disease severity

Managing the symptoms of IBD requires lifelong treatment, with severity being categorized as in remission, mild, moderate or severe¹²³. IBD severity is determined by evaluating the effect of the disease on patient symptoms and quality of life, inflammatory burden, and disease course using various grading scales^{123–125}. Although studies have found a potential sex bias in the presentation of IBD severity, the results have been contradictory¹⁰⁴. Phenotypic studies comparing the severity of IBD diagnosed in children and adults have shown that paediatric Crohn's

disease and ulcerative colitis have a more severe phenotype characterized by a more extensive, severe and aggressive disease course than those diagnosed as adults^{95,126–128}. When subdivided by sex, a Canadian population-based study derived from administrative health data with a cohort of 3,404 patients with IBD suggested that girls with IBD <21 years of age had a greater risk for hospitalization and emergency department utilization but no disparity in the rate of surgery in comparison to boys¹²⁹.

A retrospective cohort study of 989 paediatric patients demonstrated that girls with Crohn's disease had a more burdensome disease course defined by higher incidence (girls: 10 years cumulative incidence 40.2%, 95% CI 27.9–55.4%; boys: cumulative incidence 20.6%, 95% CI 15.5–27.0%) and increased risk for surgery than boys (HR 1.42, 95% CI 1.004–2.02; $P < 0.05$)^{97,130}. However, when age and sex were considered in a population-based study with 7,143 patients with pIBD, only girls with VEO ulcerative colitis had lower rates of Crohn's disease-related outpatient visits, hospitalization, emergency department utilization and surgery compared to older paediatric patients aged >6 years; no differences were noted between boys with VEO Crohn's disease and older paediatric patients (boys)⁸². Although children with Crohn's disease present predominantly with an ileocolonic disease distribution^{80,96,97,131,132}, sex has not been associated with disease localization and behaviour (fistula, stricture)^{96,97}. Girls with VEO ulcerative colitis displayed lower rates of outpatient visits (OR 0.75, 95% CI 0.63–0.94; $P < 0.0001$), whereas boys with VEO ulcerative colitis displayed a lower risk for surgery (HR 0.42, 95% CI 0.21–0.85) than older paediatric patients with IBD⁸². The hospitalization rate was higher in patients with VEO ulcerative colitis than in older paediatric patients and did not differ between the sexes⁸². Although the phenotype of ulcerative colitis in paediatric patients has been shown to have a high bias towards extensive disease and pancolitis, no difference was noted between sexes across ages^{95,96,128}. These findings suggest more limited sex bias in paediatric patients with ulcerative colitis compared to Crohn's disease. Limited population-based studies in pIBD are stratified by age and sex. Hence, there is a need for further studies to understand the underlying mechanisms associated with sex bias in patients with VEO Crohn's disease and older paediatric patients.

Sex differences in disease location have not been reported in most adult IBD studies^{106,116,133}. However, prospective Dutch studies with a total of 5,739 patients with IBD found that men with Crohn's disease had a significantly higher incidence rate of ileal disease (28%) compared to women with Crohn's disease (20%; $P < 0.01$), whereas women with Crohn's disease were more likely to have colonic disease (33%) compared to male patients (26%; $P < 0.01$)¹⁰⁶; in ulcerative colitis, proctitis was more frequent in women (10%) than in men (5%)^{106,133}.

Sex differences have also been reported in adult IBD activity, disease progression and extra-intestinal manifestations. Women with Crohn's disease are more likely to have more extra-intestinal manifestations than men¹³⁴. Yet, studies show that men had a higher rate of surgery compared to women, with a greater frequency of procedures, including stricturoplasty (6% versus 3%; $P = 0.01$), small bowel resection (16% versus 9%; $P < 0.01$) and ileocecal resection (40% versus 33%; $P < 0.01$)^{106,134}. By contrast, there were no significant differences between men and women regarding disease activity, inflammatory burden, disease progression, risk of colectomy and hospitalization for ulcerative colitis^{106,134,135}. Despite the clinical studies indicating sex differences in IBD, particularly Crohn's disease course, the underlying mechanism by which sex bias affects IBD severity has yet to be investigated.

Sex chromosomes and hormones in IBD

This influence of biological sex on innate and adaptive immune cell functions is attributed to genetic and physiological factors such as the sex chromosomes (heterologous XY for males and the homologous XX for females), reproductive organs (ovaries and testes) and sex hormones (such as oestrogens, progestogens and androgens)². Hence, the effects of sex differences on the host immune response can substantially affect an individual's susceptibility to cancers, infectious diseases, autoimmune diseases and other conditions influenced by dysregulated immune responses such as IBD⁶. Furthermore, sex differences affect known IBD risk factors such as epithelial barrier, mucosal immunity, microbiota and environmental triggers¹³⁶.

Sex chromosomes in IBD pathogenesis

In humans, the male Y chromosome contains approximately 100 genes, including genes that code for immune regulatory proteins involved in autoimmune and infectious diseases². Conversely, the human X chromosome contains more than 1,100 genes, including proteins related to pattern recognition receptors, cytokine receptors, transcription factors (elements of nuclear factor- κ B, GATA1 and forkhead box protein P3 (FOXP3)), microRNAs and long non-coding RNAs^{2,6}. Many of these genes are found to be altered in settings affected by dysregulated immunity such as IBD^{2,137}. Figure 2 visualizes the known links between sex chromosomes and hormones in IBD onset and progression across age groups.

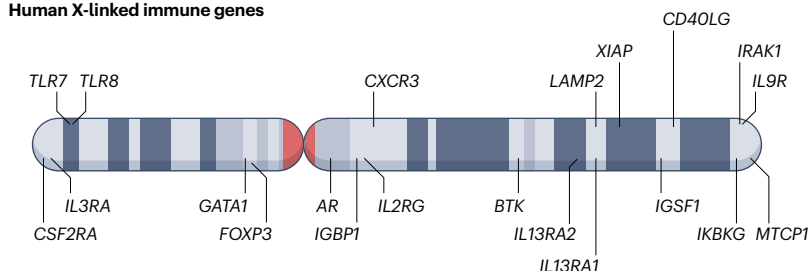
Genome-wide association studies have identified approximately 300 genetic loci associated with monogenic and polygenic IBD¹³⁸. Of these loci, several have been linked to sex chromosomes¹³⁹. Most sex chromosome loci associated with IBD discovered so far include mutations in genes related to the X chromosome, also termed X-linked genes (for example, *IKBK* (also known as *NEMO*), *FOXP3*, *XIAP*, components of NADPH oxidase complex, *BTK*, *CD40L*)¹³⁹. Mutations of X-linked genes have been associated with the onset of various clinical features of IBD; for example, mutations in *NEMO* have been linked with epithelial barrier dysfunction^{139,140}. X-linked gene disorders are monogenic and primarily affect younger patients with IBD: 63.4% of patients with VEO-IBD, 17.4% of adolescents and 10.9% of those over 18 years old¹⁴¹. IBD X-linked gene disorders affect male and female patients, although female individuals harbouring only one X-gene copy might exhibit a milder phenotype¹⁴⁰. As mentioned earlier, the Y chromosomes encode genes that are shown to regulate the immune response²; however, the exact contribution of these genes in IBD pathogenesis remains to be investigated.

Sex hormones in IBD pathogenesis

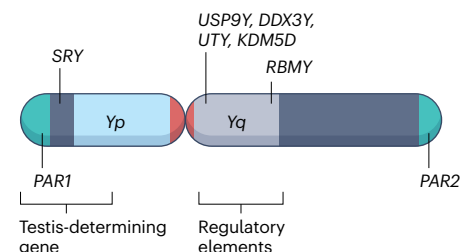
The sex hormones oestrogens, progesterone and androgens are primarily produced in the gonadal organs but also, in lesser amounts, in non-gonadal organs such as the placenta, skin, liver, heart, adipose tissue, adrenal glands, brain and intestinal mucosa^{142,143}. The concentration and type of sex hormones vary substantially throughout the lives of both female and male individuals. The primary female sex hormones include progesterone and oestrogens. There are four main types of endogenous oestrogens, including oestrone, 17 β -oestradiol, oestrinol and oestetrol^{142,144}. Of these, 17 β -oestradiol is the most potent and predominant oestrogen in females during the reproductive years. Oestrone, a weaker isoform of oestrogens, is present in more marked amounts postmenopause, whereas oestrinol and oestetrol are primarily found during pregnancy¹⁴⁴. Progesterone and oestrogens are also found in male individuals but in lower amounts: progesterone is synthesized

a Sex chromosome

Human X-linked immune genes



Human Y-linked immune genes



b Sex hormone changes by age

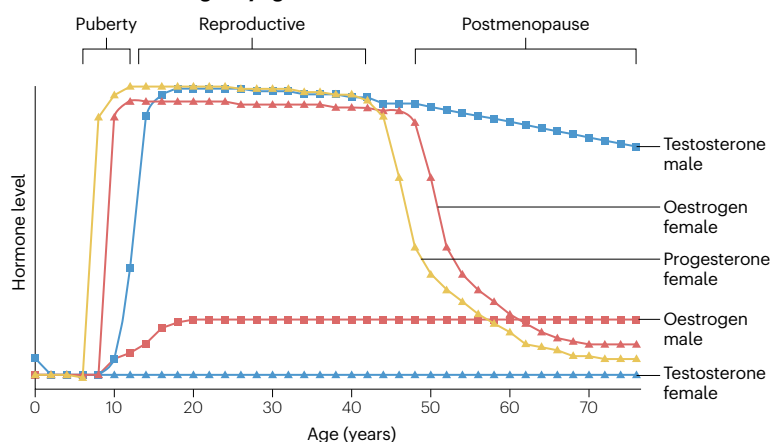
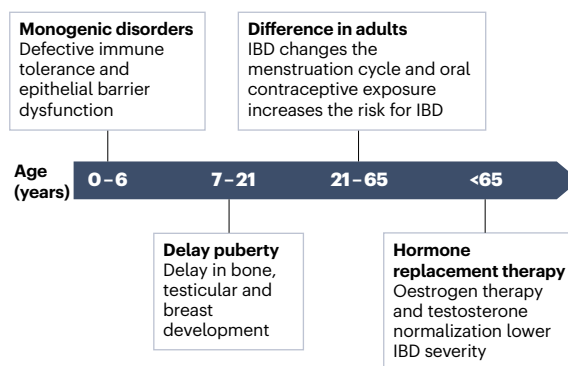


Fig. 2 | Sex chromosomes and hormones in IBD onset and progression across age groups. a. The X and Y sex chromosomes encode many genes associated with the immune system's function and regulation. Monogenic mutation of X-linked immune genes has been linked to very-early-onset inflammatory bowel disease (IBD). **b.** Sex hormones substantially fluctuate between sexes and across ages and affect IBD onset and disease severity. In IBD, early onset

c IBD clinical presentations by age



delays puberty in paediatric patients, whereas late-onset post-puberty alters the menstruation cycle in female individuals. **c.** Conversely, sex hormone therapy and supplementation in clinical and preclinical studies have shown contrasting results influenced by the sex hormone type and dose, sex hormone receptor distribution, and underlying signalling pathways. Plotted values are hypothetical and for illustrative purposes.

mainly in the testes and adrenal glands, whereas 80% of circulating oestrogens are produced within tissues by converting androgen testosterone into oestrone and 17 β -oestradiol^{145,146}. Androgens, such as testosterone and its more biologically active form, dihydrotestosterone, are the most abundant sex hormones in male persons, whereas they are produced in smaller amounts in female individuals¹⁴⁷.

Sex hormones vary substantially between sexes and across different age groups. Studies in prepubertal paediatric healthy individuals showed that girls had higher concentrations of 17 β -oestradiol than boys, whereas testosterone concentrations were comparable in both sexes^{148,149}. These trends remain similar in ages 6 to 11 years¹⁵⁰. As individuals progress through puberty (11–16 years) and transition into adulthood, sex hormone levels increase, shift between sexes and peak between 20 and 29 years^{150,151}. However, as men and women age, the concentration of sex hormones gradually decreases¹⁵¹. During menopause, women experience a drastic decrease in 17 β -oestradiol and progesterone concentrations, similar to or lower than that observed in men^{151–153}. In healthy men with average body weight, the serum concentration of testosterone, 17 β -oestradiol, oestrone and progesterone also decreased with age, although not to the same extent as in

women^{151–153}. Sex hormones have been shown to affect gut function, IBD onset and disease severity⁷⁶ (Fig. 2). An observational study evaluating the relationship between sex hormones and systemic inflammation in 72 pubertal and post-pubertal patients with Crohn's disease reported that treatment with infliximab, an anti-TNF biologic, decreased inflammation and disease activity while rapidly increasing testosterone and 17 β -oestradiol levels in boys and girls, respectively¹⁰². Changes in the concentration of sex hormones promoted sexual maturation assessed using the Tanner scale, which grades the stage of sexual development according to the development of secondary sexual characteristics for both sexes within 10 weeks of initiating infliximab treatment¹⁰². These data validated preclinical studies that suggested a regulatory role for systemic inflammation in promoting the production of sex hormones^{154–156}.

It is widely accepted that fluctuation of sex hormones related to the menstrual cycle can modulate gastrointestinal function and lead to gastrointestinal and non-gastrointestinal symptoms^{157,158}. Although menstrual changes in gastrointestinal function are common in healthy individuals, they are more pronounced in those with IBD and can result in symptoms such as constipation, nausea, vomiting, abdominal pain,

flatulence, fatigue, irritability and anxiety, amongst others^{157,158}. Further, in individuals from whom the onset of IBD occurred after puberty, the disease has been associated with changes in menstruation frequency, length of cycle, volume of flow and level of pain¹⁵⁹. It is worth noting that, in a cross-sectional study with a cohort of 1,203 patients with IBD, female patients with ulcerative colitis had a higher risk of experiencing worse symptoms compared to patients with Crohn's disease¹⁶⁰. Although still under investigation, research suggests that uterine prostaglandin might have a role in the gastrointestinal functional changes associated with menstruation¹⁶¹. Preclinical and clinical evidence demonstrated that prostaglandin E₂, which is increased during menstruation, promoted inflammation in IBD by reducing microbiota-dependent intestinal T_{reg} cells^{157,162}. Yet, several studies found that hormonal contraceptives did not necessarily aggravate symptoms during menstruation for most patients with IBD. Indeed, 75–85% of 139 patients with IBD showed no changes in symptoms, 3–20% reported improvement and only 5–7% reported worsening of symptoms¹⁵⁷. In a case–control study investigating the association between circulating androgens and the risk of IBD, the researchers noted that elevated endogenous testosterone concentration was associated with a lower risk for Crohn's disease in women but not for ulcerative colitis¹⁶³.

Sex hormone therapy in IBD pathogenesis

A meta-analysis of 20 case–control and cohort studies reported a 32% higher risk for women using oral contraceptive pills (OCPs) of developing IBD than non-OCP users¹⁶⁴. When stratified by disease subtypes, women exposed to OCP were 24% and 30% more likely to develop Crohn's disease and ulcerative colitis, respectively, compared to the non-OCP exposed women (Crohn's disease: OR 1.24, $P = 0.001$; ulcerative colitis: OR 1.25–1.30, $P < 0.05$)^{164–166}. Interestingly, studies have reported conflicting results on the relationship between exposure to OCPs and smoking in women and their effect on ulcerative colitis risk, with some reporting a lower risk and others showing a higher risk^{165–167}. A case–control study published in 2022 with a cohort of 4,932 patients with IBD matched to 29,340 healthy individuals, investigating the effect of OCP formulation, dose, length of treatment, and method of delivery on the risk of developing IBD, reported a higher risk for Crohn's disease and ulcerative colitis amongst combined OCP (COC) users (oestrogens plus progesterone) compared to non-users¹⁶⁷. These risks increased for each month of COC exposure per year, with a higher risk of 6.4% for Crohn's disease and 3.3% for ulcerative colitis compared to women not exposed to COCPs¹⁶⁷. The dose of OCP was low in contrast to standard oestrogen pills and did not influence the risk for Crohn's disease or ulcerative colitis¹⁶⁷. By contrast, the formulation of COCPs influenced the risk for Crohn's disease, with a higher risk for women using second-generation COCPs compared to newer-generation COCPs, but it did not influence the risk for ulcerative colitis¹⁶⁷. Notably, progesterone-only pills had no effect on Crohn's disease risk, whereas it was associated with a slight increase in ulcerative colitis risk (OR 1.3)¹⁶⁷. Moreover, parenteral contraception did not affect the risk for IBD¹⁶⁷, suggesting a differential function for oestrogens and progesterone in the onset of IBD subsets with a higher role for oestrogen in Crohn's disease and a prominent role for progesterone in ulcerative colitis.

One large retrospective cohort study of 108,844 women in post-menopause in the USA found that hormone replacement therapy increased the risk of ulcerative colitis in healthy women after menopause but not of Crohn's disease¹⁶⁸. Research reported that 2–3.4% of patients with endometriosis, an oestrogen-dependent condition, also had IBD, whereas 0–1% were reported in the general population¹⁶⁹.

Moreover, in a Danish nationwide study, 37,661 women hospitalized with endometriosis were more likely to develop IBD (standardized incidence ratio (SIR) 1.5; $P < 0.05$), Crohn's disease (SIR 1.6; $P < 0.05$) and ulcerative colitis (SIR 1.5; $P < 0.05$)¹⁷⁰. By contrast, hormone replacement therapy taken during the first 2 years after the onset of menopause was associated with decreased IBD activity and more conservative treatment management¹⁷¹. These studies suggest a potential role for elevated oestrogen levels in the onset of IBD, and further investigation will be needed to understand the relationship between the level of oestrogens and IBD pathogenesis.

Although the role of external hormones in women with IBD has been studied extensively, less is known about their role in men. Chronic states of low-grade inflammation have been shown to decrease testosterone levels¹⁷². A cumulative prospective study that followed 92 patients with Crohn's disease and testosterone deficiency (≤ 12.1 nmol/l) who were treated with testosterone for 7 years reported that normalizing testosterone levels led to a decrease in Crohn's disease activity index, inflammatory markers and neutrophil infiltration¹⁷³. By contrast, a cohort study conducted on 31,842 newly diagnosed men with prostate cancer who were treated with androgen deprivation therapy reported a lower risk for ulcerative colitis (HR 0.52; $P < 0.05$) and no significant risk for Crohn's disease in that population (HR 0.32)¹⁷⁴. This finding indicates a potential role for androgens in IBD pathogenesis that warrants further investigation. Despite an extensive number of studies investigating the role of oestrogens in women with IBD, the role of oestrogens and progesterone in IBD pathogenesis in men remains unclear and necessary for further investigation.

Gender dysphoria, which refers to psychological distress experienced as a result of divergence between one gender identity and sex assigned at birth, affects children, adolescents and adults¹⁷⁵. Moreover, the prevalence of self-identified transgender and non-binary (TGNB) individuals in Canada and the USA is estimated to be between 0.24% and 2.7%, of whom a large number are youth^{175–177}. Gender-affirming hormone therapy (GAHT), which includes pubertal blockers and gender-affirming hormones (oestrogen and testosterone), is commonly used to treat gender dysphoria in TGNB individuals¹⁷⁸. Pubertal blockers, such as gonadotropin-releasing hormone analogues, serve to suppress gonadotropins (follicle-stimulating hormone, luteinizing hormone) and are used by adolescents primarily to prevent testosterone or oestrogen production. Moreover, TGNB individuals can undergo feminizing hormone therapy, which utilizes oestrogen (17 β -oestrogen) along with androgen receptor blockers to prevent masculinization, whereas masculinizing hormone therapy uses testosterone¹⁷⁸. GAHT in TGNB individuals is associated with an increased risk of poor bone development, abdominopelvic pain, long QT syndrome, thromboembolism and infection, all of which are potential risk factors for IBD¹⁷⁸. A current retrospective chart review study by Hassan et al. with 23 TGNB patients with IBD reported a trend towards a higher prevalence of Crohn's disease than ulcerative colitis (59.1% versus 40.9%; $P < 0.23$). The incidence of flare was not associated with GAHT and did not differ significantly between TGNB individuals undergoing feminizing or masculinizing hormone therapy¹⁷⁹. Prospective studies with large populations will help to understand the interaction between IBD and GAHT. Moreover, they will provide insight into the effect of known IBD risk factors in the TGNB population¹⁷⁹.

Fertility in IBD course of disease

Terminologies and definitions used to discuss reproductive capacity can be ambiguous as their usage often differs across research fields

and medical settings as well as amongst various populations¹⁸⁰. The latest international glossary on infertility and fertility care defined several reproductive concepts such as fertility, fecundity, fecundability, infertility and total fertility rate¹⁸¹. In that context, fertility describes the capacity to establish a clinical pregnancy. Fecundity corresponds to the capacity to have a live birth. Fecundability is the probability of pregnancy to occur in one menstrual cycle. Infertility is associated with a failure to establish a clinical pregnancy after 12 months of regular and unprotected sexual intercourse¹⁸¹. The total fertility rate corresponds to the average number of living births per woman¹⁸¹. Parity corresponds to the number of offspring born from an individual. The fertility rate has several definitions, including the number of children per woman, the number of live births conceived in a year per 1,000 individuals of reproductive age, or the percentage of pregnant women in a population.

According to several population-based and referral centre-based studies across Western countries, IBD can marginally decrease fertility in women^{182,183}. Fertility assessment in a large population study of 27,331 women with IBD identified from the Swedish National Patient Register reported a reduced fertility rate of 1.52 ± 1.22 in women with IBD in comparison to 1.62 births per 1,000 person-years in matched control individuals ($P < 0.0001$)¹⁸³. Fertility was decreased in Crohn's disease (HR 0.88, 95% CI 0.85–0.91) and IBDU cohorts (HR 0.86, 95% CI 0.83–0.89) and marginally reduced in the ulcerative colitis cohort (HR 0.96, 95% CI 0.93–0.98) compared to the matched control cohort¹⁸³. The same study reported a reduced parity progression ratio, defined as the probability of having additional children, in women with Crohn's disease (nulliparous to parity of 1: OR 0.89, 95% CI 0.85–0.94; $P < 0.0001$) and IBDU (nulliparous to parity of 1: OR 0.87, 95% CI 0.82–0.92; $P < 0.0001$) but not with ulcerative colitis in comparison to the control group (nulliparous to parity of 1: OR 1.04, 95% CI 0.98–1.09; $P = 0.177$). Overall, these results corroborate a previous report from population-based and referral centre-based studies with a total of 946 patients, which described a decline of fertility of between 17% and 44% for patients with Crohn's disease and no significant reduction in fertility for women with ulcerative colitis¹⁸². However, the lower fertility rate in women with IBD resulted in most cases from psychosocial rather than physiological factors^{182,184,185}. Prospective international multi-centre and case–control studies reported that a higher percentage of women with IBD expressed concerns about the heritability of the condition, the risk of congenital anomalies during pregnancy, the potential teratogenic effects of medications, and the inability to financially support raising a child^{184,186–189}. As a result, some choose not to have children^{186,187}. However, extensive population-based studies have shown that the rate of involuntary infertility associated with fecundity issues amongst women was similar to that of the general population^{182–186}. Reduced fecundity in IBD can result from active gut inflammation, pelvic infection and surgical interventions¹⁸⁹. A population-based systematic review and meta-analysis that included 10,131 and 11,335 patients with Crohn's disease and ulcerative colitis, respectively, reported that patients with IBD refractory to medication or who present with complications have an increased risk (46.6% (95% CI 37.7–57.7%) for Crohn's disease and 15.6% (95% CI 12.5–19.6%) for ulcerative colitis) of undergoing abdominal surgical intervention 10 years after diagnosis¹⁹⁰. Despite the scarcity and poor quality of existing studies investigating the effect of surgery in women with IBD fertility, a study published in 2021 by Lee et al. reported a lack of evidence to support the conclusion of the role of surgery in women with IBD involuntary infertility¹⁹¹. This finding differs from several systematic reviews, meta-analyses and retrospective studies that reported an increased risk of involuntary

infertility in women with ulcerative colitis undergoing IBD-related surgeries, with infertility rates that range from 43% to 63% for ileal pouch–anal anastomosis (IPAA)^{192–194}, and that increased from 12% to 36% and 46% to 69% after restorative proctocolectomy and restorative proctocolectomy with IPAA, respectively^{193,195,196}. The aetiology of involuntary infertility in women after IBD-related surgery results from structural dysfunction such as obstruction of the fallopian tubes and pelvic adhesion^{194,191,196–198}.

Despite the scarce number of studies investigating fertility in men with IBD, several population-based and referral-centre studies reported a slight reduction of fertility in men with IBD compared to the general population. A nationwide population-based study of a cohort of 29,109 men from Sweden showed a lower fertility rate in men with IBD compared to matched control groups (1.28 ± 1.27 versus 1.35 ± 1.31 ; $P < 0.001$)¹⁹⁹. When stratified by IBD subtypes, most observational controlled studies reported reduced fertility in men with Crohn's disease of only 2–8% (HR 0.95, 95% CI 0.92–0.98) compared with matched control groups^{182,199,200}. By contrast, studies investigating fertility in men with ulcerative colitis had contradictory results. The population-based cohort study with 29,104 men with IBD and 140,901 matched individuals by Druvefors et al. showed a decreased fertility rate in men with ulcerative colitis (HR 0.93, 95% CI 0.91–0.96) and IBDU (HR 0.92, 95% CI 0.89–0.95), contrasting with previous population-based and referral-centre studies that reported no reduced fertility in men with ulcerative colitis^{182,199,200}. The minimally decreased fertility rate in men with IBD might, in part, be a consequence of psychosocial factors that lead to voluntary childlessness and reduced fecundity mediated by poor nutritional status, reduced testosterone levels, sexual dysfunction and altered semen quality in men with IBD¹⁹⁹. Possible reasons for impaired fertility in men with IBD are reduced testicular function due to zinc deficiency, decreased testosterone levels altered by the inflammation in more severe IBD activity, medical treatments, and surgical intervention^{199,201}. Two small comparative studies with 10 patients with IBD and 50 patients with ulcerative colitis reported the presence of anti-sperm antibodies, which were significantly ($P < 0.05$) elevated in the semen and serum of men with IBD and ulcerative colitis compared to control groups^{202,203}. Interestingly, anti-sperm antibodies were also found in the cervical secretion collected during ovulation and serum of women with IBD and ulcerative colitis^{202,203}. These anti-sperm antibodies might contribute to the reduced fecundability observed in men and women with IBD.

Some drugs used to treat IBD have been shown to impair male fertility. Sulfasalazine, a widely used treatment for ulcerative colitis, could lead to reversible infertility by affecting sperm count, morphology and motility¹⁹⁷. Moreover, sulfasalazine was also associated with erectile dysfunction: sulfasalazine alters the fertility of men with IBD potentially by suppressing the synthesis of sperm acrosomal membrane proteins crucial for the spermatozoa fertilization of oocytes or by increasing oxidative stress by reducing the level of redox enzymes superoxide dismutase and glutathione reductase^{184,204–206}. Either a 5-aminosalicylate can be substituted or men should be advised to discontinue the drug at least 1 month before trying to initiate pregnancy¹⁸⁴.

There are limited studies investigating fertility in men with IBD treated with immunomodulator drugs such as azathioprine, methotrexate and cyclosporine^{201,207}. Yet, several case studies studying methotrexate therapy in men with autoimmune disorders, such as rheumatoid arthritis and severe psoriasis, reported controversial findings, with some describing limited effects on male fertility. Others associated methotrexate with an increased risk for reversible infertility resulting

from erectile dysfunction and reduced sperm count, likely as a result of decreased DNA synthesis rates and subsequent inhibition of cellular proliferation owing to its folate inhibitory action^{208,209}. However, in a systematic review and meta-analysis published in 2023, using 10 studies with 268 patients with IBD, exposure to biologics (anti-TNF: infliximab, adalimumab, golimumab, certolizumab; anti-integrin: vedolizumab; JAK inhibitor: ustekinumab), azathioprine and methotrexate was not associated with impaired male fertility and adverse pregnancy outcomes²⁰⁷.

Data regarding the effect of cyclosporine on male infertility are contrasting. A few studies on a limited number of men with autoimmune infertility or following renal transplantation found no association between cyclosporine and male fertility^{210,211}. By contrast, in preclinical studies, cyclosporine treatment is linked to damaged testicular tissue, impaired sperm physiology, diminished sperm count and motility, and reduced serum testosterone concentration in rat and mouse models²¹². However, studies investigating the fertility of men with IBD using cyclosporin are lacking.

Patients with IBD frequently develop complications such as perianal abscesses, fistulas, stenosis and dysplasia-associated lesions that can require surgical intervention²¹³. A population-based study of a cohort of 310 patients by Peyrin-Biroulet et al. reported a higher risk for major abdominal surgery (HR 1.6, 95% CI 1.02–2.4; $P < 0.04$) for men with Crohn's disease²¹⁴. Moreover, results from a prospective Dutch cohort study with 2,118 patients with Crohn's disease and 1,236 patients with ulcerative colitis showed that men with Crohn's disease had a higher prevalence of undergoing small bowel and ileocecal resection (16% and 40%, respectively) than women with Crohn's disease (9% and 33%, respectively; men versus women $P < 0.01$ for both complications)¹⁰⁶. Similarly, male sex was associated with increased risk for colectomy in patients with ulcerative colitis (HR 2.1; 95% CI 1.3–3.5)²¹⁵. The most common surgical intervention for patients with IBD, that is, proctocolectomy with IPAA, has been linked with sexual dysfunction in men. A large longitudinal study with a cohort of 1,454 patients with chronic ulcerative colitis that underwent IPAA reported that 1% of men developed sexual dysfunction 1-year post-surgery, and this number doubled 10 years following surgery²¹⁶. A meta-analysis of 43 observational studies reported a 3.6% increase in the prevalence of sexual dysfunction post-IPAA in male fecundity^{213,217}. Although several other surgical interventions, such as end-ileostomy, ileo-rectal anastomosis, ileo-anal anastomosis without a pouch, colectomy and Koch pouch ileostomy, can also serve as potential therapeutic strategies in IBD, studies of their effect on men fertility are limited and will require further research.

A successful pregnancy requires maternal immune regulation or modulation to develop a tolerance to the semi-allogenic fetus. The concentration of progesterone and oestrogens during pregnancy gradually peaks in the third trimester of pregnancy²¹⁸. Moreover, their levels are markedly higher than measured throughout the reproductive cycle and have been shown to mediate the immunological changes occurring during pregnancy^{218,219}. In healthy pregnant individuals, elevated progesterone promotes the differentiation of anti-inflammatory CD4⁺ T helper 2 (T_H2) cells and immune suppressive T_{reg} cells while reducing natural killer cell activity^{219,220}. The skewed T_H2 response is associated with the secretion of anti-inflammatory cytokines such as IL-4, IL-5 and IL-10, which dampen the release of inflammatory cytokines, IFN γ and TNF, which are known to promote IBD pathogenesis²¹⁹. Moreover, this T_H2 environment correlates with increased production of blocking antibodies, lower effector antibodies and potentially increased

T_{reg} cell activity²¹⁹. Oestrogens are thought to potentially affect all cells of the innate and adaptive immune system and have been shown in preclinical and clinical studies to exert a plethora of physiological and pathophysiological functions that can be pro-inflammatory or anti-inflammatory²²¹. During pregnancy, circulating oestrogens can increase up to 60 times their non-pregnant levels, with oestriol accounting for 90% of all produced oestrogens^{218,222}. Oestrogens, specifically oestriol, are thought to have a similar effect as progesterone during pregnancy^{218,223,224}. Oestrogens might promote a T_H2 cell immune signature, increase antibody levels and improve immune tolerance by potentially boosting the number and function of T_{reg} cells^{222,224}. Therefore, this has led experts to speculate that pregnancy might improve pathology manifestations in patients with IBD. Yet, clinical studies investigating pregnancy in IBD have yielded conflicting results. Several studies reported fewer flares and a lower rate of relapse in women with Crohn's disease and ulcerative colitis measured at 3 and 10 years following pregnancy compared to pre-pregnancy rates^{76,225}. Moreover, in pregnant individuals with quiescent IBD, discontinuing anti-TNF therapy seems safe and does not lead to a relapse of either Crohn's disease or ulcerative colitis²²⁵. Most clinical practice guidelines and consensus statements consider that biologic agents can be utilized safely by women with IBD during the first stage of pregnancy and advise discontinuation of the treatment between 22 and 26 weeks of gestation. However, some reports recommend prolonging the therapy with biologic agents until 30–32 weeks of gestation or throughout pregnancy for patients with active disease or at a high risk of relapse²²⁶. By contrast, one study reported no significant differences in the progression of Crohn's disease during and after pregnancy compared to non-pregnant individuals. However, women with ulcerative colitis had a higher risk of experiencing a relapse during and after pregnancy than non-pregnant patients²²⁵. It is worth noting that the progression of IBD activity during pregnancy might be linked to disease activity during conception. Research indicates that patients who conceived during active disease were twice as likely to experience a flare-up during pregnancy as those who conceived during remission^{76,225}.

Mechanisms underlying sex bias in IBD

In addition to reproductive organs, sex hormones influence a variety of tissues, including the gastrointestinal tract and numerous haematopoietic and non-haematopoietic cells through sex hormone receptors such as oestrogen receptor isoforms (ER α , ER β and G protein-coupled oestrogen receptors (GPER; also known as GPR30)), progesterone receptor isoforms (PR-A, PR-B and PR-C) and androgen receptor isoforms^{143,227}. Beyond the most common receptor isoforms, various alternative splice variants of genes encoding sex hormone receptors have been identified in different tissues. However, the functional role of these variants is still under investigation^{228,229}. Sex hormone receptors on the cell membrane can regulate multiple signal transduction pathways and interact with growth factor receptors to enhance specific cellular processes through non-genomic pathways²³⁰. Therefore, sex-specific differences are mediated by the complexity of sex hormone receptor signalling pathways. Differences in those pathways complicate interpretations of the role of sex hormones and their receptors in physiological and pathophysiological conditions. Currently, a growing body of literature has proposed a relationship between IBD and sex hormones, particularly oestrogens, and their receptors with IBD pathophysiology^{231–235} (Fig. 3). All three oestrogen receptors have been reported in fibroblasts, vascular smooth muscle cells, endothelium of arterioles, enteric neurons and epithelial cells of the gastrointestinal mucosa^{234,236,237}.

Sex hormone signalling mechanism

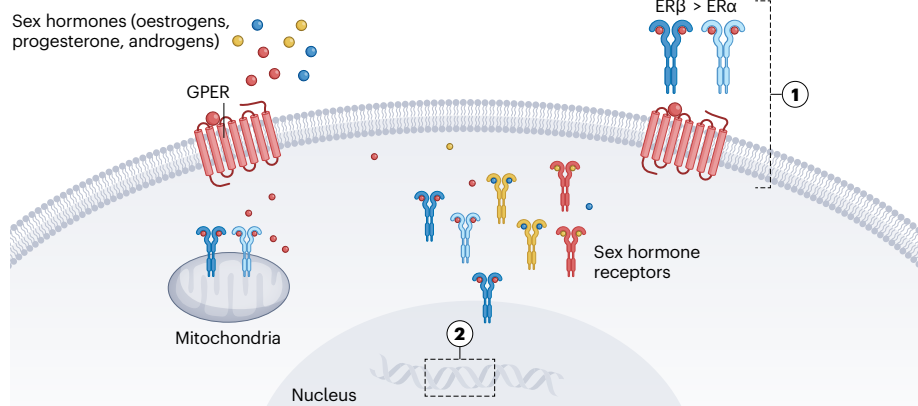
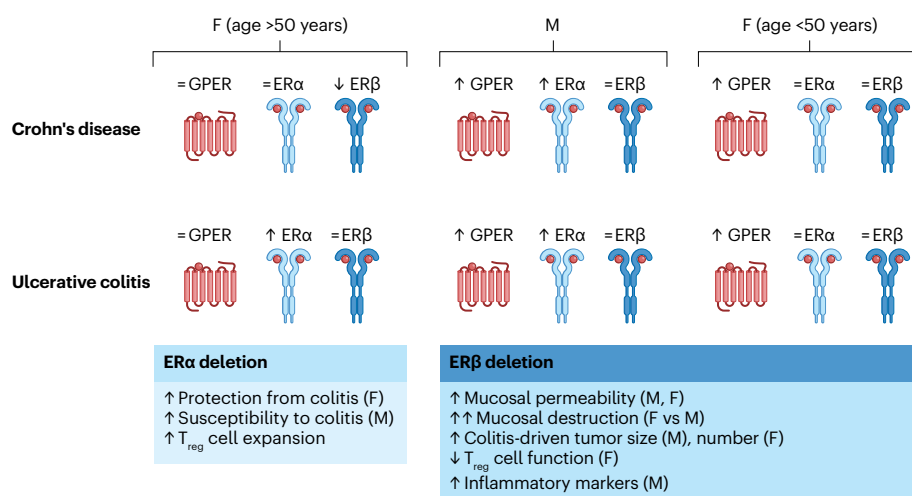
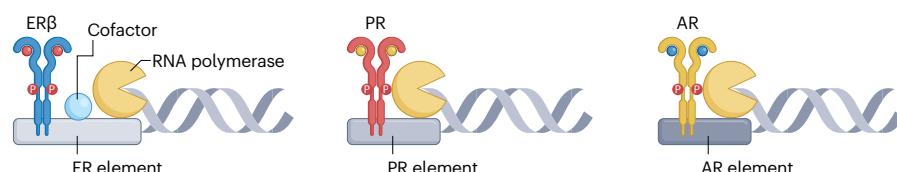


Fig. 3 | Role of dysregulated ERs in the sex discrepancy of IBD severity. Sex hormones affect the gastrointestinal tract through their interaction with the nuclear receptor family, oestrogen receptors (ERα and ERβ), G protein-coupled oestrogen receptors (GPER), progesterone receptors (PR) and androgen receptors (AR). Sex hormone receptors can function as transcription factors or form multiprotein complexes to initiate or suppress gene expression and promote epigenetic modification. ERβ is the most abundantly expressed oestrogen receptor in healthy individual intestinal tissue. However, inflammatory bowel disease (IBD) alters the expression of oestrogen receptors differentially between sexes in an age-dependent manner for female individuals. Moreover, lack of ERα or ERβ leads to contrasting outcomes in colitis susceptibility between male (M) and female (F) individuals in clinical and preclinical studies. T_{reg} cell, regulatory T cell.

1 Profile of sex hormone receptors in Crohn's disease and ulcerative colitis



2 Sex hormone receptor target genes



Biological effects of oestrogen in mouse models of colitis

A mouse study investigating dextran sulfate sodium (DSS)-mediated colitis found worsened histological outcomes in several male mice strains compared to their female counterparts, demonstrating a biological influence of sex in this setting, although results are conflicting between studies²³⁸ (Box 1). Interestingly, the effects of oestrogen supplementation were differentially associated with the experimental model of colitis utilized^{236,239,240}. In female C57Bl/6 DSS mice, treatment with a supraphysiological concentration of 17β-oestradiol, similar to those achieved during pregnancy, increased the disease severity and histological score compared with untreated mice with DSS colitis²³⁶.

By contrast, another study reported that the C57Bl/6 female mice with colitis had reduced disease activity index, macroscopic, and histological scores and lower TNF concentration than C57Bl/6 male mice with DSS-mediated colitis²³⁹. In the same study, 17β-oestradiol treatment in the ovariectomized female mice further improved female resistance to DSS-induced colitis without affecting the orchietomized male mice²³⁹. Further supporting these findings, treatment with a supraphysiological concentration of 17β-oestradiol reduced the macroscopic and histological scores in C57Bl/6 female mice with dinitrobenzene sulfonic acid-induced colitis²³⁶. This observation was accompanied by reduced neutrophilia and decreased levels of IFNγ and

IL-13, resulting in reduced colitis severity in the female dinitrobenzene sulfonic acid-treated mice²³⁶. Similarly, in the 2,4,6-trinitrobenzene sulfonic acid (TNBS)-induced colitis C57Bl/6 mice model, 17 β -oestradiol treatment of the ovariectomized females reduced colitis severity resulting from decreased pro-inflammatory cytokine expression²⁴⁰. Consistent with these findings, 17 β -oestradiol treatment diminished the histological score in *Helicobacter hepaticus*-induced mouse colitis, HLA-B27 transgenic rats and acetic acid-induced mice colitis²³⁹. Although outcomes in males are consistently worse than in females, 17 β -oestradiol treatment in ICR male mice with colitis-associated colon carcinogenesis promoted the expression of genes associated with the mucosal barrier and tight junctions along with reduced cytokine gene expression²⁴¹. In addition to being model dependent, the discrepancies resulting from 17 β -oestradiol treatment might also result from variations in 17 β -oestradiol doses administered and might point towards the need to define physiologically relevant oestrogen doses for the study of IBD.

Oestrogen action is mediated through three oestrogen receptor isoforms, namely ER α , ER β and GPER, and several ER α spliced variants, including ER α 36 and ER α 46 (ref. 242). Although all three oestrogen receptor isoforms are expressed concomitantly in the intestinal tissue, ER β is the most abundantly expressed isoform in the colon²³⁴. Despite the contrasting results, several studies have suggested a central role for dysregulated oestrogen receptor expression and signalling in the pathophysiology of IBD^{242–245}. Compared to healthy individuals, reduced mRNA levels of *Esr2* have been noted in a single-centre study with 73 patients with ulcerative colitis or active Crohn's disease. Interestingly, when segregated by sex and age, no significant changes in ER β expression were detected in men with ulcerative colitis or Crohn's disease, whereas the expression of ER β was lower in women patients over 50 years of age with Crohn's disease but not in those with ulcerative colitis²⁴². In preclinical studies, the level of *Esr2* mRNA was reduced, concomitant with increased intestinal permeability, in male and female colitis animal models (IL-10-deficient mice and HLA-B27 rats) before the onset of colitis²⁴⁵. Deletion of *Esr2* was associated with sex differences in DSS-mediated colitis, including marked mucosal destruction in female mice only²⁴⁶. Furthermore, in the absence of ER β , the expression of inflammatory markers increased most notably in male mice, along with an increasing number of colitis-driven tumours in males and tumour sizes in females²⁴⁷. Beneficial responses were uncovered following activation of *Esr2* in male DSS mice, promoting the transcription of *Nlrp6*, encoding NOD-, LRR- and pyrin domain-containing 6 (NLRP6), resulting in pro-inflammatory cytokine IL-8 secretion, antimicrobial activity, mucus production and increased autophagy²⁴⁸. Furthermore, the ER β -NLRP6 complex inhibited IL-1 α and TNF production, promoted intestinal epithelial cell survival, and reduced mitochondrial oxidative stress caused by DSS-mediated colitis, reducing mitochondrial damage²⁴⁸.

Elevated ER α and GPER protein levels are reported in intestinal tissues from 73 patients with Crohn's disease ($n = 31$) and ulcerative colitis ($n = 42$) regardless of sex and age²⁴². When segregated by sex, men with Crohn's disease or ulcerative colitis had increased levels of ER α and GPER protein expression compared to healthy individuals, whereas oestrogen receptor expression was age-dependent in women with IBD²⁴². In women under 50 years, the expression of GPER was elevated in ulcerative colitis compared to healthy individuals, whereas no significant changes were observed in the expression of ER α in both ulcerative colitis and Crohn's disease²⁴². Conversely, ER α protein expression was elevated in women over 50 years of age with ulcerative colitis but not in

those with Crohn's disease, whereas GPER protein expression remained unaltered in both ulcerative colitis and Crohn's disease²⁴². Activation of GPER with the specific agonist G-1 and 17 β -oestradiol improved the resistance of mice to TNBS-induced colitis, which was mediated through the activation of the ERK1 and ERK2 signalling pathway²⁴⁹. Although *Esr2* deletion induced mucosal damage in female DSS mice, deletion of *Esr1* resulted in protection from colitis and reduced disease severity in several mouse models; these models included the ovariectomized *H. hepaticus*-induced colitis model, T cell-mediated colitis in recombination-activating gene (RAG)-null mice and intact DSS female mice^{246,250}. Unlike females, depletion of *Esr1* in male mice enhanced male susceptibility to DSS-mediated colitis²⁴⁶. The discrepancy in disease severity between male and female mice in the absence of *Esr1* was associated with the differential expression of oestrogen-responsive genes *Socs3*, *Fos* and *Ctsd* between sexes²⁴⁶. In the absence of *Esr1*, *Ctsd* gene expression was reduced in females with DSS-mediated colitis, whereas *Fos* was elevated in males but not in females. In models where *Esr1* was depleted, researchers found elevated *Socs3* gene expression in male and female DSS mice, with no significant sex-specific differences²⁴⁶.

In contrast to the extensive research investigating the relationship between oestrogens and IBD, very few studies have explored the effect of progesterone and androgens on IBD progression and chronicity. However, some studies have shown that testosterone and progesterone might be protective against IBD^{163,251} (Box 1). A clinical study with a cohort of 83 and 91 patients with Crohn's disease and ulcerative colitis, respectively, found that women with increased testosterone levels had a lower risk of developing Crohn's disease compared to matched controls but there was no correlation with ulcerative colitis¹⁶³. A preclinical study reported a lack of effect for testosterone treatment in orchiectomized mice with DSS-induced colitis²³⁹. Treatment with progesterone was associated with reduced inflammatory markers and disease severity in male rats with TNBS-induced colitis²⁵¹.

Box 1 | Sex hormones in preclinical IBD studies

17 β -Oestradiol (E2) supplementation

- In female mice, the effect of E2 is dependent on the model, dose and receptor and can reduce or worsen colitis severity
- In male rats, E2 enhanced inflammation, gene expression of mucus, and tight junctions and restored mucosa impermeability

Progesterone (P4) and testosterone supplementation

- In female organoids, P4 reduces inflammation and improves the mucosa barrier integrity and function
- In males, P4 reduces inflammation and colitis severity. Testosterone promotes intestinal stem cell proliferation in male mice

Sex hormones and microbiota

- Gut microbiota can regulate the level of E2, P4 and testosterone. Inversely, E2 and P4 can alter microbiota diversity and abundance in mice
- Female mice display greater gut microbiota diversity than male mice

Sex bias in gut immune function in IBD

As the largest immune organ of the body, the gut houses both innate and adaptive immune cells involved in maintaining gut mucosal homeostasis²⁵². Therefore, any dysregulation of the immune compartment in the gut can lead to pathology such as IBD^{136,253}. It is well documented that most immune cells express multiple sex hormone receptors, hinting at a potential modulatory role for sex hormones on differences in the immune response between males and females in pathological conditions such as IBD²⁵⁴.

IBD is associated with a dysregulated innate immune response that leads to increased expression of pattern recognition receptors sensing receptors such as Toll-like receptor 2 (TLR2), TLR4, CD40 and CC-chemokine receptor 7 (CCR7)²⁵⁵. As a result, these receptors promote the recruitment of neutrophils and the production of inflammatory cytokines such as TNF, IL-1 β , IL-6 and IL-18 (ref. 255). Androgen and oestrogen response elements have been identified in the promoters of innate immune genes, several of which are part of the IFN, JAK2 and STAT3 signalling pathways, which are known to have an important role in IBD pathogenesis^{256–258}. Men have higher levels of TLR4 expression and more robust inflammatory responses to bacterial endotoxin lipopolysaccharide in macrophages and neutrophils than women^{2,259}. Conversely, testosterone-sensitized macrophages display reduced TLR4 and TNF²⁶⁰.

Crohn's disease is considered to evolve from a predominantly T_H1 cell response due to a high number of activated CD4⁺ T cells that secrete elevated levels of IL-2 and INF γ as well as a T_H17 cell response that causes high production of IL-23, which induces the secretion of IL-17A and INF γ while inhibiting T_{reg} cell proliferation^{261,262}. By contrast, ulcerative colitis has primarily been associated with a non-conventional T_H2 cell response resulting in the elevated secretion of IL-5 and IL-13 by mucosal CD4⁺ T cells, which might impair intestinal barrier function²⁵⁵. Oestrogen receptors are expressed in all CD4⁺ T cell subsets, including T_H1, T_H2, T_H17, T_{reg} and CD8⁺ cells²⁵⁴. Although the classical nuclear progesterone receptor is not present in lymphocytes, membrane progesterone receptor- α (mPR α) and mPR β have been identified in CD4⁺ and CD8⁺ T cells²⁶³. Additionally, mPR α protein expression on CD8⁺ T cells is sensitive to elevated progesterone levels and upregulated during the luteal phase of the menstrual cycle, whereas mPRs on CD4⁺ remain unchanged²⁶³. Mature T cells do not express androgen receptors, whereas their expression has been described in immature double-positive (CD4⁺CD8⁺) and double-negative (CD4⁺CD8⁻) T cell subsets, indicating a potential role for androgens in T cell development and maturation²⁵⁴.

Patients with IBD display increased ER α and decreased ER β expression levels in their T cells compared to healthy patients²⁶⁴. In patients with active IBD, T cells and T_{reg} cells expressing ER β were found to be low compared to healthy controls and patients in remission, whereas no significant difference was observed in T cell expression^{264,265}. The percentage of circulating T cells and CD4⁺ T cells expressing ER β was inversely correlated with disease activity and plasma concentration of inflammatory markers such as IL-6 and c-reactive protein in patients with Crohn's disease and ulcerative colitis^{264,265}. Moreover, T cells exposed to IL-6 expressed decreased levels of ER β ²⁶⁴. Interestingly, a small clinical study with a cohort of 19 patients with IBD undergoing anti-TNF treatment found that the peripheral blood T cells from patients responsive to anti-TNF treatment expressed higher protein levels of ER β than patients with IBD unresponsive to anti-TNF therapy ($P < 0.01$)²⁶⁴. By contrast, no correlation was found between ER α anti-TNF therapy and pro-inflammatory cytokines²⁶⁴. Together, these

studies support a role for T cells expressing ER β in Crohn's disease and ulcerative colitis disease progression.

Preclinical studies have shown that, although naive T cells respond to oestrogen through ER α and ER β , mature T_{reg} cells mainly respond to oestrogen through ER β ²⁴³. ER β activation has been shown to promote the expression of FOXP3 and the secretion of anti-inflammatory cytokines such as IL-10 and TGFB in T_{reg} cells²⁴³. Dysregulation in the ER β signalling pathway in a spontaneous Crohn's disease-like ileitis SAMP–YitFC mouse model promoted intestinal inflammation in the female mice in contrast to the male mice²⁴³. Deletion of *Esr2* in these SAMP–YitFC mice resulted in mRNA overexpression of the glucocorticoid-responsive transcription factor *Tsc22d3*, which inhibited T_{reg} cell suppressive function in mature female T_{reg} cells and ultimately exacerbated intestinal inflammation^{243,244}. In a chronic DSS mouse colitis model, the exclusive activation of *Esr2* could restore the population of CD4⁺ T cells expressing ER β and T_{reg} cells while also inhibiting the infiltration of effector CD4⁺ and CD8⁺ T cells in the colon mucosa²⁶⁵. Interestingly, the deletion of *Esr1* from the effector T cell population limited T cell-mediated colitis in the female *Rag^{-/-}* mouse model²⁶². More specifically, the deletion of *Esr1* in effector T cells promoted the expansion of T_{reg} cells by enhancing the protein level of FOXP3 and T_{reg} cell-associated genes while suppressing the expression of *Foxp3* suppressor genes such as *Tcf7* and *Satb1* (ref. 262). Consequently, T_{reg} cell expansion limited effector T cell activation and expansion, reduced inflammatory cytokines such as INF γ and IL-17A production, and ultimately alleviated colitis²⁶². These studies highlight the significance of maintaining a balance between oestrogen receptors in IBD to preserve intestinal mucosa tolerance in the gut. Nonetheless, further investigation is necessary to understand the underlying mechanisms that initiate the distinct activation and uneven expression of oestrogen receptors in T cells in patients with IBD. Moreover, it is essential to also investigate the role of progesterone and androgen receptors in immune cells in the context of IBD.

Sex bias on intestinal epithelium functions

Impaired epithelial barrier function resulting from elevated inflammatory cytokines is one of the hallmarks of IBD pathology²⁶⁶. The sex hormone receptors, including ER α , ER β , GPER and progesterone receptor, have been reported to be expressed in epithelial cells, whereas androgen receptors have only been reported on mesenchymal cells in the gut^{231,249,267}. In animal studies, inhibition of testosterone in male rats increased mucosal resistance to inflammation and promoted intestinal resistance to injury²⁶⁸. By contrast, in the absence of ER β , the mouse colon presented pre-pathogenic features such as abnormal histology and disrupted tight junctions associated with IBD¹⁴³. The deletion of *Esr2* led to IBD-like features such as hyperproliferation, loss of progenitor cell differentiation, decreased apoptosis and increased shedding from differentiated cells²⁶⁹. Female and male rats treated with oestrogen displayed increased occludin and junctional adhesion molecule A, improving intestinal barrier function and resistance to inflammation^{268,270}. Ulcerative colitis female patient-derived organoids treated with oestrogen increased the expression of tight junction proteins occludin, zonulin 1, claudin 1 and claudin 2 and improved the epithelial cell barrier function²⁷¹. Moreover, oestrogen treatment of organoids derived from women with ulcerative colitis decreased the production of inflammatory cytokines, reduced endoplasmic reticulum stress, induced epithelial stem cell proliferation, and promoted gut mucosal wound healing²⁷¹. Male HLA-B27 transgenic rats treated with 17 β -oestradiol or administered with ER β agonist inhibited the KCNQ1 channels and chloride

secretion, restoring mucosal impermeability and reversing chronic diarrhoea²⁷² (Box 1). Moreover, mucus-producing cell lines HT29-MTX treated with oestrogen enhanced mucin contents and promoted mucus viscosity and elasticity⁷⁶. Similarly, progesterone had a protective role in the intestinal epithelial barrier function by upregulating the tight junction proteins occludin, claudin 1 and ZO-1 and inhibiting chloride secretion^{76,271}. In addition, progesterone has been shown to reduce the secretion of inflammatory cytokines and alleviate endoplasmic reticulum stress in organoids derived from women with ulcerative colitis²⁷¹. Although the role of androgens in regulating the intestinal epithelial barrier is poorly understood, a study has suggested that androgen hormones in male mice can promote intestinal epithelial stem cell proliferation while inhibiting their differentiation by downregulating the bone morphogenetic protein signalling pathways²⁶⁷.

Sex differences and gut microbiota

Gut microbiota can regulate host physiology through many actions that preserve the intestinal mucosa architecture, influence the epithelial barrier function, produce energy, protect against pathogenic microbiota colonization, and modulate the development and function of the immune system^{273,274}. On the other hand, dysbiosis and microbiota-produced carcinogenic metabolites and toxins can drive pathologies such as IBD²⁷⁵.

Dysbiosis has been shown to have a crucial role in IBD pathogenesis, with well-documented marked differences in the mucosa-associated microbial communities between individuals with and without IBD^{276–280}. Patients with Crohn's disease often display a reduced relative abundance of Firmicutes and Bacteroidetes concomitant with an elevated relative abundance of Proteobacteria compared to patients with ulcerative colitis and individuals without IBD²⁸⁰. The complexity of studying the microbiota in IBD rests heavily on the heterogeneity of the microbiota community between individuals and even within the same individual throughout the day²⁸¹. This diversity in the composition of gut microbiota is shaped by genetics, the host immune system, intestinal mucosa barrier function and environmental factors, including geographical location, age, health, lifestyle, diet and sex hormones^{273,282} (Box 1). Although controversial, some preclinical and clinical studies reported sex-specific differences in gut microbiota composition^{282,283}. Moreover, microbiota communities have been shown in mice to metabolize sex hormones, consequently regulating the effect of sex hormones on gut mucosal functions^{282,284–286}. The diversity of the gut microbiota differs between sexes at the genus and species level²⁸⁷. At the genus level, *Bilophila* and *Blautia*, two genera associated with IBD, were markedly more abundant in faecal samples from women compared to those of men, whereas *Veillonella* and *Methanobrevibacter* were more predominant in samples from men²⁸⁷. At the species level, a higher abundance of *Bacteroides cacae* was uncovered in faecal samples from women, whereas *Bacteroides plebeius* and *Coprococcus catus* were elevated in samples from men²⁸⁷. Moreover, marked differences in the microbial composition and diversity at the phylum level have been noted between the sexes in 89 different strains of mice, which is influenced by several factors²⁸².

Several studies have described the importance of sex hormones in establishing sex-mediated differences in the gut microbiota^{285,286,288} (Box 1). However, the gut microbiota difference mediated by sex is dependent on age^{282,288}. In a non-obese diabetic mouse model, gut microbiota diversity and composition were not different between male and female mice before puberty²⁸⁹. Yet, puberty was associated with lower microbiota diversity in the gut of males compared to females²⁸⁹.

Male non-obese diabetic (NOD) mouse model of type 1 diabetes mice housed in specific pathogen conditions were more protected than female NOD mice in the same conditions²⁸⁵. Conversely, the microbiota of germ-free male NOD mice was similar to that of germ-free female NOD mice²⁸⁵. Castration of male NOD mice resulted in a microbiota that resembled that of female NOD mice microbiota²⁸⁹. These findings from mouse studies suggest that androgens might act as one of the drivers of sex differences in gut microbiota composition^{285,286,288}. When investigating the effect of oestrogens on gut microbiota, a clinical study reported a lack of correlation between the concentration of 17 β -oestradiol in women and sex-mediated differences in gut microbiota²⁸⁸. Yet, oestrogens have been shown to inhibit the growth of *Escherichia coli*, *B. fragilis* and *Fusobacterium nucleatum* in rat intestines²⁹⁰. By contrast, a clinical study with a cohort of 91 pregnant women found that progesterone increased the abundance of *Proteobacteria* and *Actinobacteria* and decreased the abundance of *Faecalibacterium*^{76,291}. In addition, progesterone increased the abundance of *Bifidobacterium* in humans and mice during pregnancy²⁹². All these findings suggest a role for sex hormones in microbiota diversity and abundance²⁸⁸. However, confounding factors, such as genetic background, age and environment, can regulate gut microbiota, complexifying the interpretation of data²⁸². Thus, subsequent investigations are required to decipher the underlying mechanisms that regulate the role of the microbiota in sex bias in IBD.

The underlying mechanisms that govern sex-mediated influences in IBD are complex and involve a network of multiple players, such as sex hormones, sex hormone receptors, microbiota, immune cells and epithelial cells, whose cumulative influences from each other determine the host gastrointestinal resistance to insults and IBD outcome. Our current knowledge indicates that oestrogen supplementation reduced disease severity in male and conditionally in female mice by promoting the expansion of T_{reg} cells, production of anti-inflammatory cytokines, intestinal mucosa integrity and barrier function^{270,271,293}. These effects seem to require the presence of *Esr2*, whose expression is susceptible to inflammatory factors and the female reproductive cycle^{247,294}. Similarly, although scarce, studies report a protective role for both progesterone and testosterone as they promote intestinal mucosal integrity (Box 1). Notably, although regulated by the microbiota, testosterone can alter intestinal epithelial cell differentiation²⁶⁷.

Inclusion of sex in IBD research

Sex as a biological variable must be accounted for in the research design, participant selection, sample size, data collection, statistical analysis, reporting, and dissemination in preclinical and clinical studies^{16,295,296}. Poorly designed biomedical research costs lives and resources. In basic biomedical research, failing to account for biological sex might lead to inaccurate conclusions, whereas in clinical studies, such shortcomings might delay timely diagnosis and treatment and harm patients. Moreover, sex-combined analyses when sex differences exist might lead to erroneous conclusions or masking effects between the sexes. Thus, an appropriate analysis of sex should be paramount in IBD research because of growing evidence in this field pertaining to sex differences in almost all aspects of this disease globally.

Despite the emphasis in current years to appropriately use the terminology of sex and gender in research, the implementation of these terms into practice remains, for the most part, limited and approximate. The probable reluctance or confusion associated with the utilization of these terms is likely due to poor understanding by researchers of what these terms entail. Therefore, to conduct research

that investigates biological sex and gender differences, it is crucial to explicitly clarify how sex and gender are defined while formulating the research question.

Research design to investigate sex effect in IBD

Two research strategies, namely method and factorial designs, are used to investigate the effect of sex as a biological variable²⁹⁶. The method design explores the difference in the main effect on both sexes separately²⁹⁵. This approach has the advantage of using a smaller sample size to characterize the study effect and sample size in both sexes. However, the method design fails to assess the interaction between the main effect and the variable sex^{295,297}. By contrast, data disaggregated by sex can be integrated into a factorial design to statistically test the interaction between variables and whether sexes respond differently to an intervention^{295,296}. In addition, considering factorial designs in experimental studies helps to reduce the sample size required for sex-based comparisons²⁹⁸. Monte Carlo simulation and ANOVAs can be suitable statistical analyses to compare the correlation or difference between groups in the factorial research design^{295,296}.

Participant selection in clinical studies

Defining confounding factors during research design will help to guide data collection²⁹⁵. Studies should account for confounding factors such as sex, gender, age, smoking status, age of disease onset, environment, reproductive stage and cycle, and hormonal therapy to reduce or eliminate the effect of these covariables on sex-mediated interactions in IBD and to expand research validity and generalizability¹⁶. Overlooking confounding factors might lead to inaccurate data interpretation and conclusion.

Physiological and behavioural sex differences attributed to prenatal and perinatal activation have been demonstrated before puberty²⁹⁹. At puberty, the reproductive maturation phase involves sex-specific changes in the central nervous system, sex hormone production, and secondary sexual characteristics such as the growth of testicles and facial hair, breast formation, and the menstrual cycle³⁰⁰. Hence, studies should define the inclusion and exclusion criteria associated with reproductive maturation stages¹⁶. In longitudinal research, accounting for reproductive history should be an important consideration because of its potential to influence the cohort under consideration. For example, data acquisition might be affected if a patient gets pregnant during a longitudinal study. Moreover, pregnancy-related substantial changes in oestradiol and progesterone levels might serve as a modulator for the association between other factors^{301,302}.

Model selection for basic and translational research studies

Animal models are crucial tools to investigate sex-dependent processes experimentally, examining genetic and sex hormone modifications such as gonadectomy with sex hormone treatments, transgenic animal models lacking sex hormone receptors, and four-core genotype (FCG) mouse models^{2,303}. Cell lines are also a potential alternative although they are poor models for the study of sex hormone-mediated changes—most cell lines used in basic biomedical research are derived from one sex only. In addition, cell-line use raises the issue associated with a sample size of one patient from whom they are derived and limits the ability to distinguish between differences linked to sex, environmental and genetic factors¹⁶.

The use of explant biopsy samples, patient-derived cells and peripheral blood mononuclear cell models, such as organoids, can help to mitigate the limitation associated with using cell lines as they

mirror the pathophysiological changes observed in IBD and reflect diversity in the patient population^{304,305}. However, these model systems do not recapitulate the complete gut microstructure as they lack the complex interaction between the microbiota, epithelium, immune and stroma components, all known to differ between sexes^{2,76,304,305}. Moreover, despite being an excellent source for cells to investigate the immune response in IBD research, the immune cell phenotypes in peripheral blood mononuclear cells vary markedly from that of the intestinal mucosa, highlighting the need for caution when interpreting the data³⁰⁴. The underlying cellular and molecular mechanisms associated with sex hormones and receptors in IBD pathology are still unclear². Clinical studies have investigated the effect of sex chromosomes and hormones by comparing the immune response of healthy individuals of both sexes to individuals with Klinefelter or Turner syndrome. Klinefelter and Turner syndrome are two inherited disorders that result from either supplementary X chromosome (47, XXY) or X deletion (45, X or 45, XO)^{2,137}. These instances also raise an important point that biological sex exists in a spectrum.

The link between sex chromosomes, gonad-type development and sex steroid hormone production might pose a challenge to discern the influence of sex chromosomes and steroid hormones (and their potential interaction) depending on the intended aim or goal of a particular study. In preclinical studies, an excellent mouse model, the FCG model, can serve to investigate the distinctive effect of sex chromosomes and hormones³⁰³. FCG mice are produced by either deletion of the *Sry* gene from the Y chromosome (XY⁻), which leads to a female phenotype, or by insertion of the *Sry* transgene onto an autosomal chromosome (XX) or XY⁻ FCG mice to produce the XX^{Sry} and XY^{Sry} FCG mice that are phenotypically male³⁰³. FCG mice can be supplemented with sex hormones to investigate the interaction between sex hormones and the gonadal type or genetic component (that is, sex chromosomes) in the preclinical model of IBD.

Sample size

Sample size is an important consideration for robustly assessing sex differences because of its importance in determining substantial differences between the experimental groups under consideration and this should reflect population prevalence. Researchers should aim to balance research validity, which requires narrowing subject selection, generality and large population selection, by defining the appropriate sample size data¹⁶. The statistical power necessary to detect sex differences is affected by the prevalence of the pathology in each sex, the outcome, and the strength of the interaction between sex and the main effect³⁰⁶. Thus, it is crucial to plan studies with enough statistical power to detect the difference in the main effect between sexes³⁰⁶. This planning, though, might be particularly complicated for the study of gender as it might be challenging to conduct prospective or randomized control trial studies that include gender-diverse individuals with a sample size large enough to detect gender differences owing to the small size of the known gender-diverse individuals in the general population and the relatively small number of IBD cases worldwide^{307,308}.

Analysis and interpretation of sex differences in IBD

New government mandates mean that more studies have begun to address the role of sex as a biological variable and that increasing numbers of researchers will be collecting and analysing data by sex^{16,309}. A simple search of 'sex differences in IBD' using PubMed (as of 30 June 2024) shows that the majority (>78%) of articles have been published in the past 10 years. Growing evidence suggests that there is a need for

thoughtful consideration of sex-related differences in IBD research and lessons can be learned from other disciplines^{16,309–311}. Including sex as a biological variable not only has the potential to promote discovery and expand the relevance of biomedical research but it also provides a foundation of knowledge on which to build future advances^{16,311,312}. To this end, we must utilize best practices to include sex considerations in study design, data collection, analysis and interpretation to establish the rigour and reproducibility needed to inform clinical decision-making and improve outcomes³¹². Without careful analysis and interpretation of the effects of sex differences on IBD, the pursuit of sex-difference research, despite a mandate from funding agencies and scientific journals, will result in a body of literature of contradiction. There are many inconsistent and contradictory reports on the presence, magnitude and importance of sex differences in various aspects of IBD, including prevalence and incidence³¹³, disease onset and progression during growth and ageing⁹⁷, and treatment outcomes^{134,314–316}. Some of them might be due to poor study design, analysis and interpretation as well as their subsequent perpetuation in review articles.

Criteria that need consideration are the intentional and accurate use of language and terminology when referring to sex and sex steroid hormones, an in-depth sex analysis incorporated into study designs and reporting, and the inclusion of sex considerations in the interpretation of study results^{317,318}. The use of proper terminology is essential in the study, analysis and interpretation of sex-related differences in IBD, which are often overlooked. There are many examples of the imprecise use of even common terminology in the field of sex difference-based studies in IBD research in current literature. For example, the first deliberate investigation on sex-related differences in IBD was a study published in 1999 in which the body's hydration and the distribution of the body's water compartments were assessed compared with healthy individuals³¹⁹. Unfortunately, in this study, the term 'gender' was used incorrectly to describe biological sex. In addition, 'gender' has incorrectly been used in many studies with a focus on biological sex^{96,242,320}, including a preclinical animal study published in 2013 on metabolic profiling³²¹. Notably, many of these articles have been published in journals in the field of gastroenterology and IBD^{322–324}, which is concerning. Moreover, in many articles, 'gender' and 'sex' have been used interchangeably^{106,322,323}. It is counterintuitive to describe sex differences using erroneous terminology. In addition, IBD is a complex, multifactorial and multidimensional disease influenced by a variety of factors, including age, genetics, sex, gender, environment and geography³²⁵, which are difficult to capture comprehensively in any study. Thus, a description of 'limitations' is paramount in all studies and analyses pertaining to various aspects of IBD. Moreover, there is a need for caution in using the information gleaned from these studies and for further research to replicate the findings, which also requires consideration. For example, sex steroid hormones are an integral component in sex-related differences in physiology and pathophysiology at large, including various aspects of IBD^{76,326}. However, the modulatory effect of sex steroids on IBD might include their action on immune and non-immune cells of the gastrointestinal tract as well as the gut microbiota and its potential interplay with the host, because all of these factors contribute to sex-related differences in various aspects of IBD^{327–330}. The relationship between sex steroids, gut microbiota and IBD could be multifaceted³³¹. Thus, analysis and interpretation of data from an *in vivo* study on the modulatory role of sex steroids on immune cells of the gastrointestinal tract or microbiota alone are bound to have limitations and require a cautious approach in reporting sex differences in IBD. Together, these considerations,

if implemented, are expected to reduce the perpetuation of incorrect or unclear information and provide better practices in the study of sex-related differences in IBD in the future.

Similarly, the imprecise, mixed use of terminology describing the male and the female sex steroid hormones (such as androgens, oestrogens and progesterone) are often found in scientific literature attempting to address the effects of sex differences. For example, oestrogen versus oestradiol and androgen versus testosterone have interchangeably been used in analysis and interpretation in many studies^{76,332}. These terminologies are also used in a mixed format (for example, oestrogen and testosterone)^{76,332}. It is important to note that oestrogens and androgens are a group of compounds, whereas 17 β -oestradiol and testosterone are the most common hormones in each of these respective groups. Different forms of oestrogens and androgens differ in their potencies, and their concentration levels do not follow a uniform pattern during ageing^{333,334}. In addition, a simple description of sex steroid hormone levels is not sufficient because it is their free, biologically available form (and not a carrier protein-bound, biologically unavailable form) that is more important in their actions on cells and tissues^{335,336}. Additionally, the response of sex steroid hormones is regulated in different target cells by co-regulators^{337,338} that need consideration in analysis and interpretation or should at least be included in the limitations of the study. Progesterone concentration levels must also be considered.

Researchers from many disciplines are beginning to appreciate the importance of considering sex differences in the design, analysis and interpretation of their studies^{309,310,312,318}, factors still underappreciated in the study of IBD. It is important to determine if there is a sex difference in the expression of the IBD trait in female and male patients. Then, one must consider whether the sex difference is attributable to the actions of sex steroids, sex chromosomes or other factors. To answer the potential role of sex steroids requires knowledge of how to appropriately assess and manipulate the hormonal condition of the patients in the study³¹⁸. The analysis and interpretation of sex differences in IBD must pre-specify tests of sex differences to reduce the occurrence of a type I error, to account for confounding by factors associated with sex, and to rule out instances when sex differences were tested as post hoc exploratory analyses, and these analyses must interpret the apparent sex differences in the light of biological plausibility^{16,309}. Subgroup analyses from studies thoughtfully designed to query sex differences can provide sound evidence of the benefits of sex-based treatments to target IBD.

It is crucial to understand that there are different types of sex differences such as qualitative (that is, the variables are not related to each other in the same way across sexes) and quantitative sex differences^{16,309}. Qualitative sex-difference analyses should be conducted independently by sex, whereas in quantitative sex differences, the average or mean of a dependent variable is different for female and male individuals, and analyses should be conducted with sex as a factor^{16,309}.

When defining key sex differences in translational and clinical research, studies must complete more than simple comparisons between male and female persons to avoid overemphasizing sex differences¹⁶. The source of the observed sex differences is an important consideration, including environmental, genetic, hormonal and, potentially, other factors². Moreover, adjusting the possible intersecting and confounding factors, such as age, is essential to consider because sex differences in IBD seem to have an age-dependent relationship and sex distribution of IBD epidemiology differs between different geographical regions^{4,83,98}. In longitudinal studies, how sex differences evolve over

time during different stages in life and how observed differences might vary by factors such as age, ethnicity and socio-economic status are important to examine³³⁹. In addition, a lack of statistical power to test the interactions associated with sex differences can lead to overinterpretations of the findings in studies investigating sex and gender differences in IBD¹⁹⁷. Thus, investigators should address the ability of the study to limit bias, eliminate confounding factors, and discuss surprise subgroup findings discovered through post-analysis^{16,197}.

Big data encompasses a systematic analysis of large volumes of complex data to uncover trends, patterns and correlations that generate valuable new insights and have the potential to help address the sex-specific knowledge gap that currently exists in biomedical and health-care research^{340,341}. In current years, the wide adoption of electronic health records combined with clinical information and public health records and the growing development and adoption of multi-omics approaches have provided the scientific community with large repositories of structured, semi-structured and unstructured data³⁴². Systemic and integrative analysis of big data using advanced machine learning and artificial intelligence techniques could help to design effective patient-centric health care and disease management and provide real-world data for observational studies to extract real-world evidence to improve medical outcomes and population health³⁴⁰. The use of big data in IBD research could help to identify sex and/or gender differences in the disease risk predictors, manifestation, diagnosis, treatment and prognosis, alleviating our knowledge gap associated with sex and gender disparity in IBD³⁴⁰.

Reporting sex-specific differences in IBD

Finally, the dissemination of findings from experimental, clinical and epidemiological studies also requires equal attention and adherence to the current mandate by different funding organizations and scientific journals in reporting sex differences in IBD. A few specifics that need consideration are reporting the sex of the study participants (even in single-sex studies), sex distribution of the study participants (cells, animals or humans), how information on sex was obtained, disaggregation of results by sex, whether observed sex differences are of practical importance, and reporting all results – positive, negative and inconclusive¹¹.

Conclusion

Accumulating evidence shows differences in IBD pathogenesis, epidemiology, clinical course and disease outcomes associated with sex^{1–9}. As several studies demonstrate that sex-mediated factors can differentially regulate IBD risk factors such as genetic predisposition, dysregulated immune response, impaired mucosal function, environmental exposure and intestinal dysbiosis, it is crucial to integrate sex as a biological variable in IBD research¹³⁶. Thus, the appropriate integration of sex in research design and data analysis will reduce the inconsistencies observed between current studies, promote new insights into IBD pathophysiology and IBD management, extend the relevance of IBD research to all, and ultimately improve patient care.

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Author contributions

D.M.T. and H.K.A. researched data for the article. C.N.B., D.M.T., J.-E.G. and H.K.A. contributed substantially to discussion of the content. D.M.T., C.N.B., S.M. and H.K.A. wrote the article. All authors reviewed and/or edited the manuscript before submission.

Competing interests

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